User Guide for SAP Cloud Infrastructure

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SAP Cloud Infrastructure | SAP Customers and Partners

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SSH Key Pairs

Enable secure access to your server instances. A key pair consists of a private key, which you keep safe on your local machine, and a public key, which you upload to your cloud account.

Uploading an SSH Public Key

Enable secure SSH access to your server instances by uploading your public SSH key.

Prerequisites

- You have generated an SSH key pair on your local machine.
- You have access to your public SSH key in OpenSSH format.

Example

- macOS and Linux: `~/.ssh/id_rsa.pub` or `~/.ssh/id_ed25519.pub`
- Windows: `C:\Users\<`*`your name`*`>\.ssh\id_rsa.pub` or `C:\Users\<`*`your name`*`>\.ssh.id_ed25519.pub`

Procedure

1. From the user menu in the top header bar, choose **Key Pairs**.

   The key pairs UI page opens, showing any existing keys.
2. Choose **Create new**.

   A dialog box opens.
3. Enter your name.
4. Paste your public key in OpenSSH format.
5. Choose **Save**.

Results

- Your public key is now registered. When creating a new server instance, you can select it from the **Key Pair** dropdown to enable secure SSH access using the matching private key.
- The activity is recorded as an event in the audit log.

Next Steps

- Creating Server Instances
- Launching Instances

Related Information

Audit

Creating a user-data File to Inject Multiple SSH Public Keys

Create a `user-data.yaml` file to inject multiple SSH public keys into a server instance during creation. The keys are added using `cloud-init`.

Prerequisites

- You have access to the SSH public keys you want to inject.

Procedure

1. On your local machine, create a file named `user-data.yaml`.
2. Add the following content to the file. Replace the example keys with your actual SSH public keys:

   Code Syntax

   ```
   #cloud-config
   ssh_authorized_keys:
     - "ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAABAQC0...key1"
     - "ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAABAQC1...key2"
     - "ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAABAQC2...key3"
   ```

   Each key must be on a single line. Use correct indentation.
3. Save the file.

Next Steps

- Creating Server Instances
- Launching Instances

Deleting an Uploaded SSH Public Key

Remove an existing SSH public key from your registered key pairs to prevent it from being used for new server instances.

Context

Deleting an uploaded SSH public key from the dashboard removes it from the list of available key pairs for creating or launching new server instances. This action does not delete the corresponding private key stored on your local machine. The private key remains on your device unless you remove it manually.

Additionally, any existing server instances that were created with this key pair remain accessible using the corresponding private key unless you manually remove the key from the server's `~/.ssh/authorized_keys` file.

Procedure

1. From the user menu in the top header bar, choose **Key Pairs**.

   The key pairs UI page opens, showing any existing keys.
2. From the gear dropdown menu for the key pair, choose **Delete**.
3. Confirm the dialog box.

Results

- The public key is removed from your registered key pairs. It no longer appears as an option when creating or launching new server instances.
- The activity is recorded as an event in the audit log.

Related Information

Audit

Flavors

A flavor is a predefined hardware configuration for a server, defining the size of a virtual server to be launched, including its compute, memory, and storage capacities.

Flavor Categories and Naming Conventions

Get an overview of flavor naming conventions, including how workload type, CPU, memory, and hardware generation are represented.

Flavors Catalog

Get an overview of available flavors.

Viewing Flavors

View detailed information about available flavors.

Flavor Categories and Naming Conventions

Get an overview of flavor naming conventions, including how workload type, CPU, memory, and hardware generation are represented.

Flavor Categories

Workload Type	Prefix	Description	Flavors
Compute-Optimized	c	Compute-heavy tasks At least 1 vCPU for every 2GiB RAM	Virtual Machine Flavors
General	g	Balanced compute and memory	

Workload Type	Prefix	Description	Flavors
		1 to 2 vCPUs for every 4GiB RAM	
Memory-Optimized	m	Memory-intensive tasks More than 4GiB RAM per vCPU	
SAP HANA-Optimized	hana	Optimized for in-memory SAP HANA workloads	[SAP HANA Flavors]

Flavor Naming Conventions

Flavor names follow a general naming convention to help you identify key characteristics like workload type, CPU, and memory. These conventions are for guidance only and not guaranteed:

*<workload-type>*_c*<vCPUs>*_m*<RAM>*_v*<version>*

Components Explained

Component	Description
<workload-type>	Indicates the flavor's intended workload: • c = Compute-Optimized • g = General • m = Memory-Optimized • hana = SAP HANA-Optimized
_c*<vCPUs>*	Number of vCPUs or cores.
_m*<RAM>*	Amount of RAM in GiB.
_v*<version>* (Optional)	Hardware generation indicator.

Related Information

[Flavors Catalog]

Flavors Catalog

Get an overview of available flavors.

Virtual Machine Flavors

Flavor	vCPU	vRAM (GiB)	Root Disk Size (GiB)	Workload Type	Supersedes
g_c1_m2	1	2	64	General	m1.xsmallcpuhdd
g_c1_m3	1	3	64	General	
g_c1_m4	1	4	64	General	

Flavor	vCPU	vRAM (GiB)	Root Disk Size (GiB)	Workload Type	Supersedes
c_c2_m2	2	2	64	Compute-Optimized	m1.small, m1.smallhdd
g_c2_m4	2	4	64	General	m1.xsmall, x1.xsmallhdd
g_c2_m8	2	8	64	General	m1.xmedium
c_c4_m4	4	4	64	Compute-Optimized	m1.medium
g_c4_m8	4	8	64	General	m1.large
g_c4_m16	4	16	64	General	m1.xlarge
m_c4_m32	4	32	64	Memory-Optimized	
m_c4_m64	4	64	64	Memory-Optimized	m2.large
g_c6_m24	6	24	64	General	
g_c8_m16	8	16	64	General	m2.xlarge
g_c8_m32	8	32	64	General	m1.2xlarge
m_c8_m64	8	64	64	Memory-Optimized	m2.4xlarge
m_c8_m128	8	128	64	Memory-Optimized	
m_c8_m256	8	256	64	Memory-Optimized	
g_c12_m24	12	24	64	General	
g_c12_m48	12	48	64	General	
c_c16_m16	16	16	64	Compute-Optimized	m1.xlarge_cpu
g_c16_m32	16	32	64	General	m1.2xlargecpu
g_c16_m64	16	64	64	General	m1.4xlarge
m_c16_m128	16	128	64	Memory-Optimized	m2.8xlarge
m_c16_m256	16	256	64	Memory-Optimized	
m_c16_m512	16	512	64	Memory-Optimized	
m_c20_m160	20	160	64	Memory-Optimized	
m_c24_m192	24	192	64	Memory-Optimized	
g_c32_m128	32	128	64	General	
m_c32_m256	32	256	64	Memory-Optimized	
m_c32_m512	32	512	64	Memory-Optimized	
g_c48_m192	48	192	64	General	
m_c48_m512	48	512	64	Memory-Optimized	

Flavor	vCPU	vRAM (GiB)	Root Disk Size (GiB)	Workload Type	Supersedes
g_c64_m256	64	256	64	General	m5.16xlarge
m_c64_m512	64	512	64	Memory-Optimized	
g_c128_m512	128	512	64	General	m5.32xlarge
m_c128_m1000	128	1,000	64	Memory-Optimized	

SAP HANA Flavors

Flavor	vCPU	vRAM (GiB)	Sockets (NUMA)	CPU Model	CPU Architecture	advanced SAPS	SAPS
hana_c24_m365	24	365	1/2	Intel Xeon Platinum 8260	Cascade Lake	8,300	45,650
hana_c48_m729	48	729	1	Intel Xeon Platinum 8260	Cascade Lake	11,000	60,500
hana_c96_m1458	96	1,458	2	Intel Xeon Platinum 8260	Cascade Lake	21,800	119,900
hana_c144_m2188	144	2,188	3	Intel Xeon Platinum 8260	Cascade Lake	32,300	177,650
hana_c192_m2917	192	2,917	4	Intel Xeon Platinum 8260	Cascade Lake	43,800	240,900
hana_c288_m4377	288	4,377	6	Intel Xeon Platinum 8260	Cascade Lake	62,100	341,550
hana_c384_m5835	384	5,835	8	Intel Xeon Platinum 8260	Cascade Lake	82,200	452,100
hana_c30_m480_v2	30	480	1/2	Intel Xeon Gold 6448H	Sapphire Rapids	15,800	86,900
hana_c60_m960_v2	60	960	1	Intel Xeon Gold 6448H	Sapphire Rapids	18,800	103,400
hana_c120_m1920_v2	120	1,920	2	Intel Xeon Gold 6448H	Sapphire Rapids	39,200	215,600
hana_c180_m2880_v2	180	2,880	3	Intel Xeon Gold 6448H	Sapphire Rapids	53,900	296,450
hana_c240_m3840_v2	240	3,840	4	Intel Xeon Gold 6448H	Sapphire Rapids	81,400	447,700

Flavor	vCPU	vRAM (GiB)	Sockets (NUMA)	CPU Model	CPU Architecture	advanced SAPS	SAPS
hana_c480_m7680_v2	480	7,680	8	Intel Xeon Gold 8454H	Sapphire Rapids	129,900	714,450
hana_c736_m15351_v2	736	15,351	8	Intel Xeon Platinum 8468H	Sapphire Rapids		

Related Information

[SAP Standard Application Benchmarks](#)

Viewing Flavors

View detailed information about available flavors.

Prerequisites

- You have the `member` role.

  For more information, see [Authorizations](#).

Procedure

Choose **Compute** > **Flavors**.

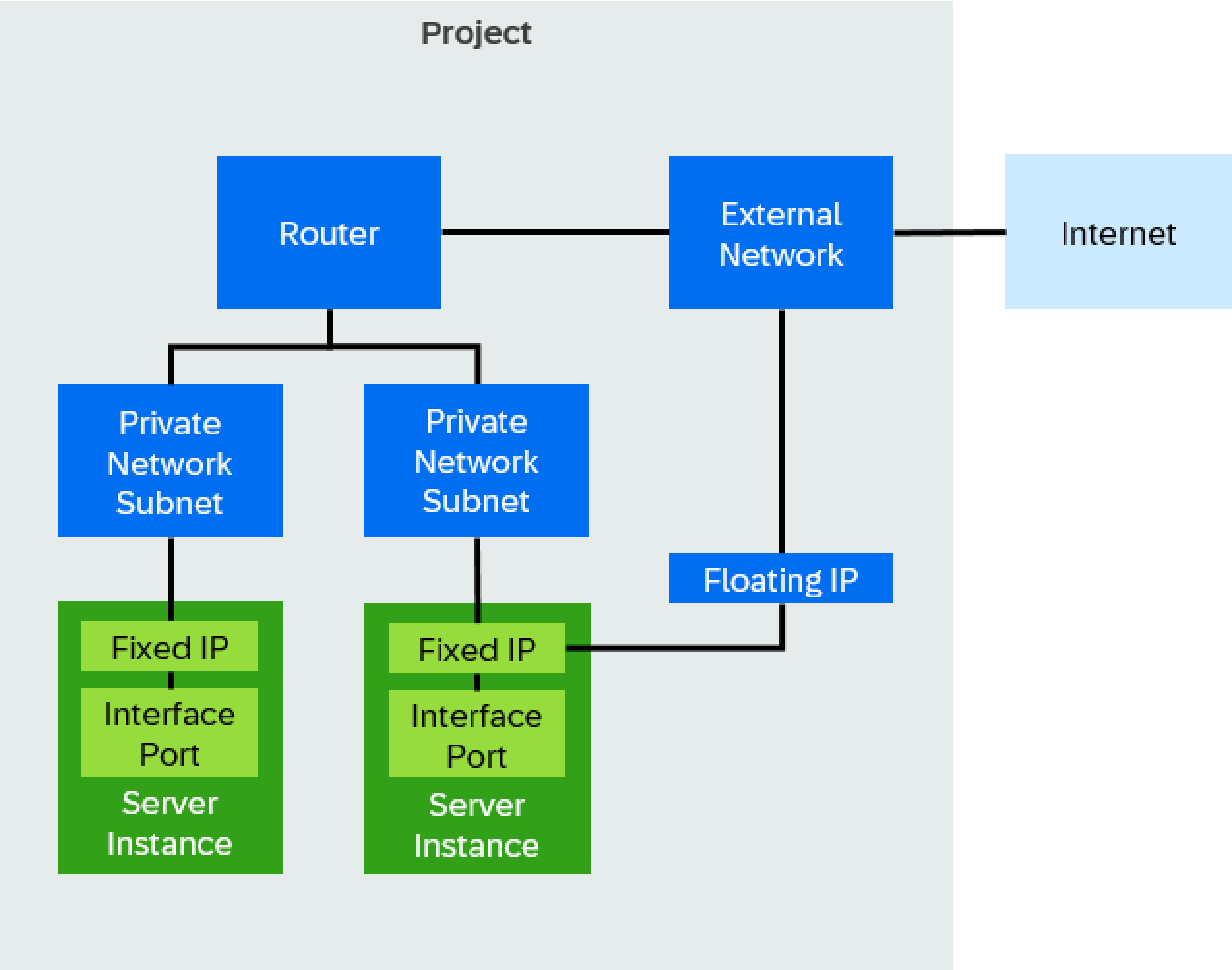
A table lists the available flavors with the following attributes:

Attribute	Description
Name	The identifier for the flavor.
VCPUs	The number of virtual CPUs allocated to the flavor.
RAM	The amount of memory available for the server instance.
Root Disk	The size of the root disk, where the operating system is installed.
Ephemeral Disk	Temporary storage that is deleted when the server instance is deleted.
Swap Disk	Additional swap space allocated for the server instance.
RX/TX Factor	The network bandwidth factor associated with the server instance.
Public	Indicates whether the flavor is available for public use.

Networking and Loadbalancing

Manage network connectivity and traffic distribution to optimize application performance and reliability.

This image is interactive. Hover over each area for a description.



Please note that image maps are not interactive in PDF outputs.

[Supported Network Connections](#)

Get an overview of supported network connections.

[Network Traffic Rules](#)

Get an overview of network traffic that is always enabled, regardless of security groups applied.

[External Networks](#)

Projects need at least one external network for an Internet connection. Without these networks, users can't access instances in the private network, leading to isolation. Domain owners create and manage external networks and their associated subnets.

[Private Networks](#)

Private networks isolate resources within a project and assign internal IP addresses for communication. Create, edit, and delete private networks, manage subnets, and share private networks with other projects.

[Routers](#)

A router connects subnets and external networks, enabling communication within your infrastructure. Manage router settings, configure routers, and view your router topology.

[Border Gateway Protocol VPN](#)

Border gateway protocol (BGP) VPNs are virtual private networks that use BGP to exchange routing information between connected networks. Manage BGP VPNs by creating them, adding or removing routers, sharing them with other projects, and deleting them when they are no longer needed.

[Fixed IPs / Ports](#)

Fixed IPs are IPv4 addresses that DHCP assigns to server instances (VMs). A fixed IP acts as a port on the private network and connects to the NIC of a server instance.

[Floating IPs](#)

Floating IPs (FIPs) are static IPs that belong to a project rather than a specific server instance. You can assign them to any server instance for SSH, RDP, or HTTP access. FIPs are created from the IPv4 pool of a subnet.

Security Groups

Manage security groups that act as virtual firewalls to control incoming (ingress) and outgoing (egress) network traffic for one or more instances. Security groups consist of security group rule sets that define the allowed network traffic for all instances within the security groups.

Load Balancers

Load balancers distribute incoming network traffic across multiple server instances to improve availability and prevent overload. Configure load balancers to manage traffic flow, monitor health, and ensure efficient resource use.

Domain Name System

Manage domain name system (DNS) zones and records.

Supported Network Connections

Get an overview of supported network connections.

At the transfer point of the direct connection in the SAP data center, the following network ports are available for single-mode fiber optic connections:

- 10GBASE-LR: Supports 10 Gbps over long-range single-mode fiber.
- 100GBASE-LR4: Supports 100 Gbps over long-range single-mode fiber.

Other port types require a separate agreement.

Network Traffic Rules

Get an overview of network traffic that is always enabled, regardless of security groups applied.

Outbound Traffic

- ICMP IPv4/IPv6
- Metadata Service (HTTP @ 169.254.169.254:80)
- DHCPv4/v6 Requests

Inbound Traffic

- ICMP IPv4/IPv6
- DHCPv4/v6 Responses

External Networks

Projects need at least one external network for an Internet connection. Without these networks, users can't access instances in the private network, leading to isolation. Domain owners create and manage external networks and their associated subnets.

Related Information

Creating Projects

Private Networks

Private networks isolate resources within a project and assign internal IP addresses for communication. Create, edit, and delete private networks, manage subnets, and share private networks with other projects.

[Creating Private Networks](#)
Create isolated private networks to enable secure internal communication and resource management within your project.
[Editing Private Networks](#)
Update the name or admin state of an existing private network.
[Deleting Private Networks](#)
Permanently delete a private network you no longer need.
[Sharing Private Networks with other Projects](#)
Share private networks with other projects using access control.
[Creating Subnets in Private Networks](#)
Create subnets in private networks to segment traffic.
[Deleting Subnets from Private Networks](#)
Permanently delete subnets from private networks when they are no longer needed or part of the network's design. This action frees up IP address space and removes unused resources.

Creating Private Networks

Create isolated private networks to enable secure internal communication and resource management within your project.

Prerequisites

- You have the `network admin` role.

  For more information, see Authorizations.

Procedure

1. Choose **Networking & Loadbalancing** > **Network & Routers**.
2. Choose the **Private Networks** tab page.
3. Choose **Create new**.
4. Enter a name.
5. Select the admin state: **UP**
6. Enter a subnet name.
7. Enter the network address and subnet mask (CIDR).

   → Recommendation

   Use the default CIDR range `10.180.0.0/16` or a subrange.

Related Information

Creating Routers

Editing Private Networks

Update the name or admin state of an existing private network.

Prerequisites

- You have the `network admin` role.

  For more information, see [Authorizations](#).

Procedure

1. Choose **Networking & Loadbalancing** > **Network & Routers**.
2. Choose the **Private Networks** tab page.
3. From the gear dropdown menu for the private network, choose **Edit**.

   A dialog box opens.
4. Enter a name.
5. Select the admin state. You have the following options:
   - UP
   - DOWN
6. Choose **Save**.

Deleting Private Networks

Permanently delete a private network you no longer need.

Prerequisites

- You have the `network admin` role.

  For more information, see [Authorizations](#).

Procedure

1. Choose **Networking & Loadbalancing** > **Network & Routers**.
2. Choose the **Private Networks** tab page.
3. From the gear dropdown menu for the private network, choose **Delete**.

Sharing Private Networks with other Projects

Share private networks with other projects using access control.

Prerequisites

- You have the `network admin` role.

For more information, see Authorizations.

- You have a private network.

  For more information, see Creating Private Networks.

Procedure

1. Choose **Networking & Loadbalancing** > **Network & Routers**.
2. Choose the **Private Networks** tab page.
3. From the gear dropdown menu for the private network, choose **Access Control**.

   A dialog box opens.
4. Choose +.
5. Select or enter the ID of the target project allowed to use the private network.
6. Choose **Add**.

Related Information

Border Gateway Protocol VPN

Creating Subnets in Private Networks

Create subnets in private networks to segment traffic.

Prerequisites

- You have the `network admin` role.

  For more information, see Authorizations.

Context

Create subnets within predefined CIDR ranges, such as the default CIDR range `10.180.0.0/16` or its subranges.

i Note

To create subnet IP ranges outside these predefined ranges, use the CLI.

Procedure

1. Choose **Networking & Loadbalancing** > **Network & Routers**.
2. Choose the **Private Networks** tab page.
3. From the gear dropdown menu for the private network, choose **Manage Subnets**.

   A dialog box opens.
4. Choose +.
5. Enter a subnet name.
6. Enter the network address and subnet mask (CIDR).

→ **Recommendation**

Use the default CIDR range `10.180.0.0/16` or a subrange.

7. Choose **Add**.

Related Information

[Command-Line Guide for SAP Cloud Infrastructure]

Deleting Subnets from Private Networks

Permanently delete subnets from private networks when they are no longer needed or part of the network's design. This action frees up IP address space and removes unused resources.

Prerequisites

- You have the `network admin` role.

  For more information, see Authorizations.

Procedure

1. Choose **Networking & Loadbalancing** > **Network & Routers**.
2. Choose the **Private Networks** tab page.
3. From the gear dropdown menu for the private network, choose **Manage Subnets**.

   A dialog box opens.
4. Choose 🗑.

Routers

A router connects subnets and external networks, enabling communication within your infrastructure. Manage router settings, configure routers, and view your router topology.

Routers Configuration Maximum
Get an overview of the configuration limits for routers in VM infrastructures.

Creating Routers
Create routers to connect multiple private network subnets to external networks. A router also enables one-to-one NAT between the private networks and external networks via floating IPs.

Editing Routers
Update the name, floating IP network, subnet, and private network subnets of an existing router.

Deleting Routers
Permanently delete a router you no longer need.

Viewing the Router Topology
Get a visual representation of the router topology.

Routers Configuration Maximum

Get an overview of the configuration limits for routers in VM infrastructures.

Resource	Maximum Size / Count	Comments
Private Networks per Router	100	Maximum number of private networks that can be connected to one router.
Floating IPs per Router	1024	Maximum number of floating IP addresses that can be assigned to a router.
Directly Accessible Private Network (DAPN) IPs / Ports per Router	1280	Maximum number of DAPN IPs/ports that can be passed through a router.
Static Routes per Router	256	Maximum number of all static IPv4/6 routes that can be assigned to a router. The router can dynamically learn additional routes through BGP VPN.

Creating Routers

Create routers to connect multiple private network subnets to external networks. A router also enables one-to-one NAT between the private networks and external networks via floating IPs.

Prerequisites

- You have the `network admin` role.

  For more information, see Authorizations.

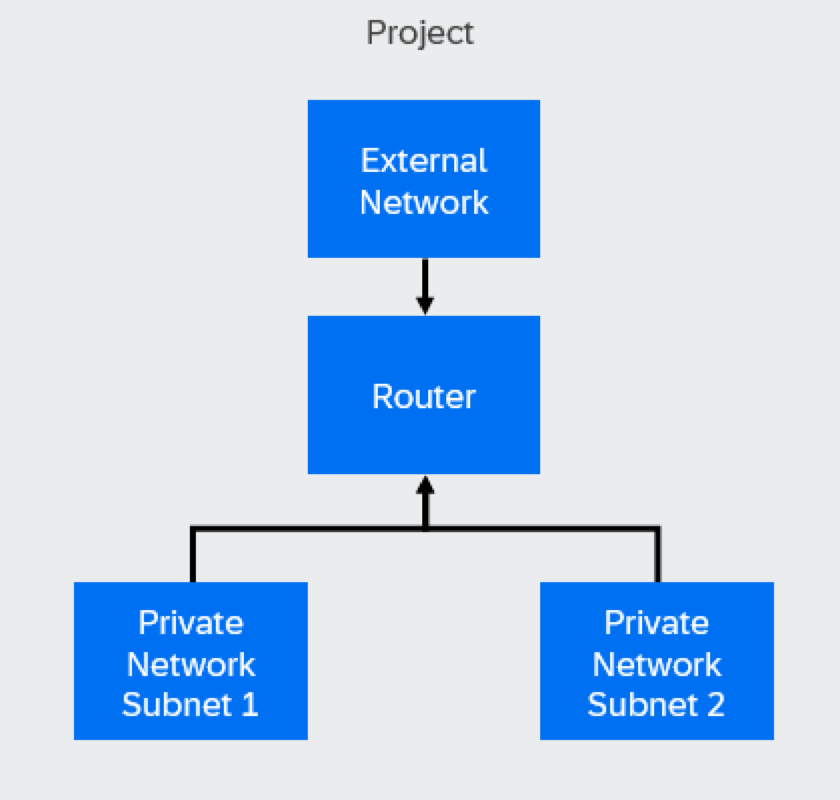
- You know and comply with the configuration limits for routers.

  For more information, see Routers Configuration Maximum.

- You have private networks.

  For more information, see Private Networks.

Context



A router connects multiple private network subnets to a single external network and enables one-to-one NAT for communication via floating IPs.

Procedure

1. Choose Networking & Loadbalancing > Network & Routers.
2. Choose the Routers tab page.
3. Choose Create new.

   A dialog box opens.
4. Enter a name.
5. **Optional:** Select the admin state. You have the following options:
   - UP
   - DOWN
6. Select the floating IP network.
7. Select the private network subnets to attach.

   You can attach private subnets to only one router at a time.
8. Choose Create.

Next Steps

To verify the router's configuration, try pinging from an instance in one private network subnet to an instance in another private network subnet.

Editing Routers

Update the name, floating IP network, subnet, and private network subnets of an existing router.

Prerequisites

- You have the `network admin` role.

  For more information, see Authorizations.

- You know and comply with the configuration limits for routers.

  For more information, see Routers Configuration Maximum.

Procedure

1. Choose **Networking & Loadbalancing** > **Network & Routers**.
2. Choose the **Routers** tab page.
3. From the gear dropdown menu for the router, choose **Edit**.

   A dialog box opens.
4. Enter a name.
5. Select the floating IP network.
6. Select the subnet.
7. Select the private network subnets.
8. Choose **Update**.

Deleting Routers

Permanently delete a router you no longer need.

Prerequisites

- The router you want to delete has no associated objects, such as routes, gateways, or ports.

- You have the `network admin` role.

  For more information, see Authorizations.

Procedure

1. Choose **Networking & Loadbalancing** > **Network & Routers**.
2. Choose the **Routers** tab page.
3. From the gear dropdown menu for the router, choose **Delete**.

Viewing the Router Topology

Get a visual representation of the router topology.

Procedure

1. Choose **Networking & Loadbalancing** > **Network & Routers**.
2. Choose the **Routers** tab page.

3. From the gear dropdown menu for the router, choose **Topology**.

Border Gateway Protocol VPN

Border gateway protocol (BGP) VPNs are virtual private networks that use BGP to exchange routing information between connected networks. Manage BGP VPNs by creating them, adding or removing routers, sharing them with other projects, and deleting them when they are no longer needed.

[Cross-Project Interconnections]

Use border gateway protocol (BGP) VPN to enable communication between virtual machines (VMs) in different projects within the same region.

[Cross-Region Interconnections]

Use border gateway protocol (BGP) VPN to enable communication between networks located in different regions by creating interconnections between BGP VPNs.

[Creating Border Gateway Protocol VPNs]

Create border gateway protocol (BGP) VPNs for your project.

[Adding Routers to Border Gateway Protocol VPNs]

Connect routers to border gateway protocol (BGP) VPNs to enable subnet announcements.

[Removing Routers from Border Gateway Protocol VPNs]

Remove the association between routers and border gateway protocol (BGP) VPNs to stop announcing subnets connected to the routers.

[Sharing Border Gateway Protocol VPNs with other Projects]

Share border gateway protocol (BGP) VPNs with other projects using access control.

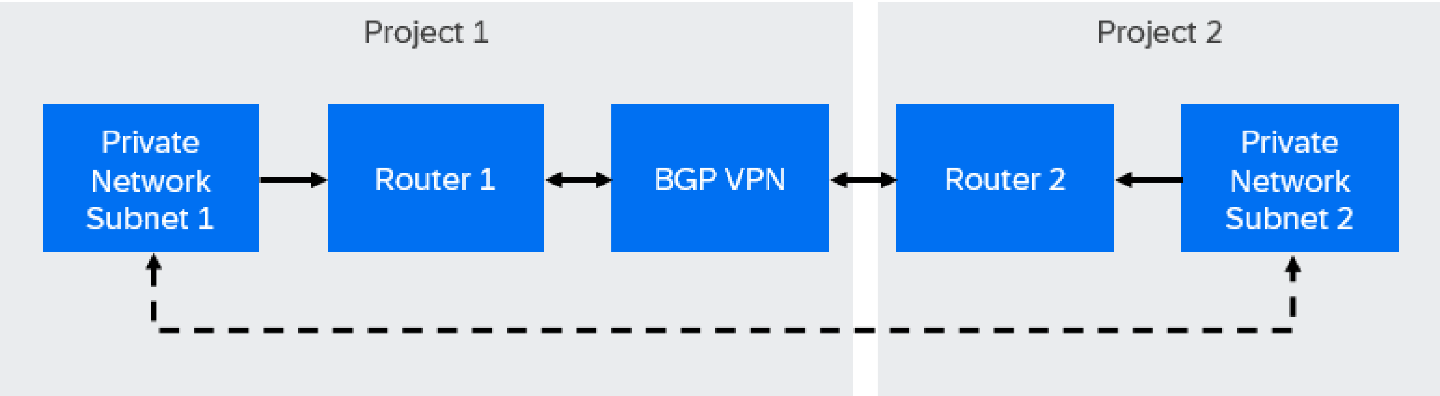
[Deleting Border Gateway Protocol VPNs]

Permanently delete a border gateway protocol (BGP) VPN you no longer need.

Cross-Project Interconnections

Use border gateway protocol (BGP) VPN to enable communication between virtual machines (VMs) in different projects within the same region.

How It Works



Each private network is connected to a router, and the BGP VPN is created in one of the projects. The BGP VPN is then shared with the second project. By establishing associations between the routers and the BGP VPN, VMs in different projects can communicate directly without using public IP addresses.

Requirements

- Two separate private networks in different projects with distinct subnets.

  For more information, see:

  - Creating Private Networks
  - Creating Subnets in Private Networks

- A router connected to each private network.

  For more information, see Creating Routers.

- Both projects must belong to the same domain.
- A BGP VPN created in one of the projects.

  For more information, see Creating Border Gateway Protocol VPNs.

- The BGP VPN shared with the second project.

  For more information, see Sharing Border Gateway Protocol VPNs with other Projects.

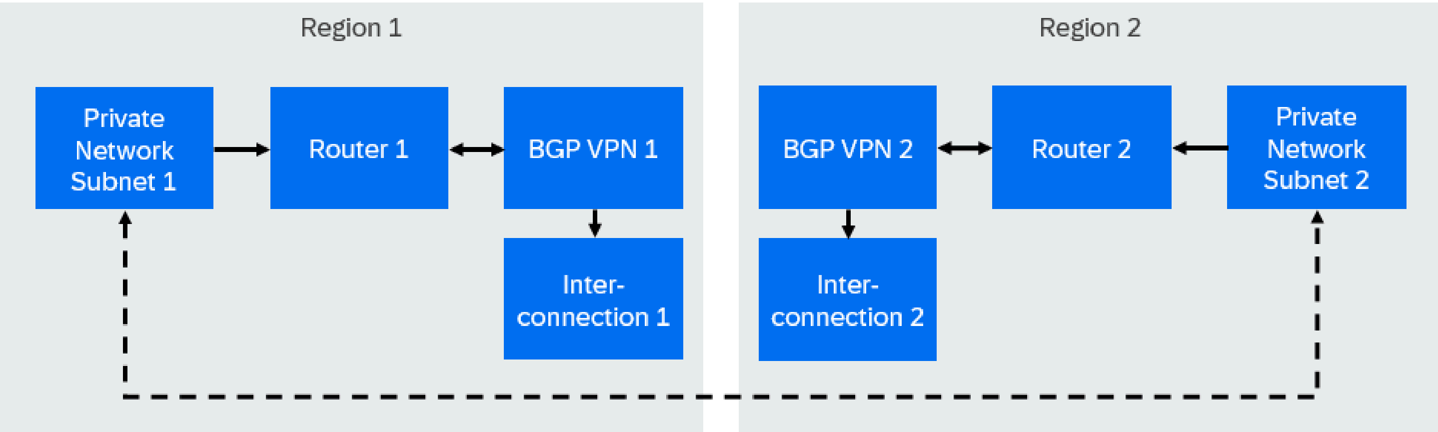
- Associations established between the routers and the BGP VPN.

  For more information, see Adding Routers to Border Gateway Protocol VPNs.

Cross-Region Interconnections

Use border gateway protocol (BGP) VPN to enable communication between networks located in different regions by creating interconnections between BGP VPNs.

How It Works



Each BGP VPN operates independently within its respective region. To establish communication between the private networks, an interconnection object is created, linking the BGP VPNs. This interconnection acts as a bridge, ensuring traffic from one region reaches the correct destination in the other region while maintaining isolation.

Requirements

- Private networks in each region with distinct, non-overlapping subnets.

  For more information, see:

  - Creating Private Networks

- Creating Subnets in Private Networks

- A router connected to each private network.

  For more information, see Creating Routers.

- Both projects must belong to the same domain.

- Two separate BGP VPNs, one in each region.

  For more information, see Creating Border Gateway Protocol VPNs.

- An interconnection object that links both BGP VPNs.

  For more information, see Required Plugins.

Creating Border Gateway Protocol VPNs

Create border gateway protocol (BGP) VPNs for your project.

Procedure

1. Choose **Networking & Loadbalancing** > **Network & Routers**.
2. Choose the **BGP VPNs** tab page.
3. Choose **New BGP VPN**.

   A dialog box opens.
4. Enter a name.
5. Choose **Save**.

Next Steps

- Adding Routers to Border Gateway Protocol VPNs
- Sharing Border Gateway Protocol VPNs with other Projects

Adding Routers to Border Gateway Protocol VPNs

Connect routers to border gateway protocol (BGP) VPNs to enable subnet announcements.

Prerequisites

- You have a router.

  For more information, see Creating Routers.

Procedure

1. Choose **Networking & Loadbalancing** > **Network & Routers**.
2. Choose the **BGP VPNs** tab page.
3. From the gear dropdown menu for the BGP VPN, choose **Manage Routers**.

   A dialog box opens.

4. Select a router.
5. Choose **Add**.

Removing Routers from Border Gateway Protocol VPNs

Remove the association between routers and border gateway protocol (BGP) VPNs to stop announcing subnets connected to the routers.

Procedure

1. Choose **Networking & Loadbalancing** > **Network & Routers**.
2. Choose the **BGP VPNs** tab page.
3. From the gear dropdown menu for the BGP VPN, choose **Manage Routers**.

   A dialog box opens.
4. Choose 🗑.
5. Choose **Confirm**.

Sharing Border Gateway Protocol VPNs with other Projects

Share border gateway protocol (BGP) VPNs with other projects using access control.

Procedure

1. Choose **Networking & Loadbalancing** > **Network & Routers**.
2. Choose the **BGP VPNs** tab page.
3. From the gear dropdown menu for the BGP VPN, choose **Access Control**.

   A dialog box opens.
4. Enter the project name or ID of the target project.
5. Choose **Add**.

Deleting Border Gateway Protocol VPNs

Permanently delete a border gateway protocol (BGP) VPN you no longer need.

Procedure

1. Choose **Networking & Loadbalancing** > **Network & Routers**.
2. Choose the **BGP VPNs** tab page.
3. From the gear dropdown menu for the BGP VPN, choose **Delete**.
4. Confirm the dialog box.

Fixed IPs / Ports

Fixed IPs are IPv4 addresses that DHCP assigns to server instances (VMs). A fixed IP acts as a port on the private network and connects to the NIC of a server instance.

Reserving Fixed IPs

Reserve fixed IP addresses to ensure consistent network connectivity for network devices or server instances (VMs) within your private network.

Editing Fixed IP Ports

Update the security group and description of an existing fixed IP port.

Editing Ports

Update the security group and description of an existing port.

Deleting Fixed IP Ports

Permanently delete a fixed IP port you no longer need.

Reserving Fixed IPs

Reserve fixed IP addresses to ensure consistent network connectivity for network devices or server instances (VMs) within your private network.

Prerequisites

- You have the `network admin` role.

  For more information, see Authorizations.

- **Optional:** You have created a security group.

  For more information, see Creating Security Groups.

Procedure

1. Choose **Networking & Loadbalancing** > **Fixed IPs / Ports**.
2. Choose **Reserve new IP**.

   A dialog box opens.
3. Select the network.
4. Select the subnet.
5. **Optional:** Enter the IP.
6. **Optional:** Select the security group.
7. **Optional:** Enter a description.
8. Choose **Save**.

Related Information

Creating Server Instances

Editing Fixed IP Ports

Update the security group and description of an existing fixed IP port.

Prerequisites

- You have the `network admin` role.

  For more information, see Authorizations.

- **Optional:** You have created a security group.

  For more information, see Security Groups.

Procedure

1. Choose **Networking & Loadbalancing** > **Fixed IPs / Ports**.
2. From the gear dropdown menu for the fixed IP port, choose **Edit**.

   A dialog box opens.
3. **Optional:** Select the security groups.
4. **Optional:** Enter the description.
5. Choose **Save**.

Editing Ports

Update the security group and description of an existing port.

Prerequisites

- You have the `network admin` role.

  For more information, see Authorizations.

Procedure

1. Choose **Networking & Loadbalancing** > **Fixed IPs / Ports**.
2. Select the **other ports** radio button.
3. From the gear dropdown menu for the port, choose **Edit**.

   A dialog box opens.
4. **Optional:** Select the security groups.
5. **Optional:** Enter the description.
6. Choose **Save**.

Deleting Fixed IP Ports

Permanently delete a fixed IP port you no longer need.

Prerequisites

- You have the `network admin` role.

  For more information, see Authorizations.

Procedure

1. Choose **Networking & Loadbalancing** > **Fixed IPs / Ports**.
2. From the gear dropdown menu for the fixed IP port, choose **Delete**.
3. Confirm the dialog box.

Floating IPs

Floating IPs (FIPs) are static IPs that belong to a project rather than a specific server instance. You can assign them to any server instance for SSH, RDP, or HTTP access. FIPs are created from the IPv4 pool of a subnet.

[Creating Floating IPs](#)

Create a public IP address that you can later attach to your server instances to enable internet access. If you specify a DNS domain, the network service automatically creates an A record for name resolution.

[Editing Floating IPs](#)

Update the description of an existing floating IP.

[Releasing Floating IPs](#)

Release unused floating IPs back to the network to free up resources.

Creating Floating IPs

Create a public IP address that you can later attach to your server instances to enable internet access. If you specify a DNS domain, the network service automatically creates an A record for name resolution.

Prerequisites

- You have the `network admin` role.

  For more information, see Authorizations.

- You have an external network.

  For more information, see External Networks.

- **Optional:** To automatically create an A record for the floating IP, you have a domain name system (DNS) zone.

  For more information, see Domain Name System.

Procedure

1. Choose **Networking & Loadbalancing** > **Floating IPs**.
2. Choose **Allocate new**.

   A dialog box opens.
3. Select the external network.
4. Select the external subnet.
5. **Optional:** Select the DNS domain.
6. **Optional:** Enter the DNS name that you want to use with the floating IP.

Ensure that the DNS name is a valid partially qualified domain name (PQDN) or fully qualified domain name (FQDN) and is no longer than 63 characters.

7. Choose **Allocate**.

Results

- The network service allocates a floating IP and associates it with the selected external network.
- If you specified a DNS domain and name, the network service creates an A record for name resolution.
- The activity is recorded as an event in the audit log.

Next Steps

Attaching Floating IPs to Server Instances

Related Information

Audit

Editing Floating IPs

Update the description of an existing floating IP.

Prerequisites

- You have the `network admin` role.

  For more information, see Authorizations.

Procedure

1. Choose **Networking & Loadbalancing** > **Floating IPs**.
2. From the gear dropdown menu for the floating IP, choose **Edit**.

   A dialog box opens.
3. Edit the description.
4. Choose **Save**.

Results

- The activity is recorded as an event in the audit log.

Related Information

Audit

Releasing Floating IPs

Release unused floating IPs back to the network to free up resources.

Prerequisites

- You have the `network admin` role.

  For more information, see Authorizations.

- **Optional:** You have detached the floating IP from any server instance.

  For more information, see Detaching Floating IPs from Server Instances.

Procedure

1. Choose **Networking & Loadbalancing** > **Floating IPs**.
2. From the gear dropdown menu for the floating IP, choose **Release**.

Results

- If the floating IP is attached to a server instance, it's automatically detached first. The released IP returns to the external network and becomes available for a new allocation.

- The activity is recorded as an event in the audit log.

Related Information

Creating Floating IPs
Audit

Security Groups

Manage security groups that act as virtual firewalls to control incoming (ingress) and outgoing (egress) network traffic for one or more instances. Security groups consist of security group rule sets that define the allowed network traffic for all instances within the security groups.

Security Groups Configuration Maximum
Get an overview of the configuration limits for security groups in VM infrastructures.

Creating Security Groups
Create security groups to manage network traffic for one or more instances.

Adding Security Group Rules to Security Groups
Security group rules specify the network traffic which is allowed for all instances within a security group.

Sharing Security Groups with other Projects
Define which target projects can use a security group. This enables reuse across multiple projects without duplicating settings and prevents target projects from modifying them.

Deleting Security Group Rules from Security Groups
Delete security rules that you no longer need from a security group.

Deleting Security Groups
Permanently delete a security group you no longer need.

Related Information

Network Traffic Rules

Security Groups Configuration Maximum

Get an overview of the configuration limits for security groups in VM infrastructures.

Resource	Maximum Size / Count
Security Group Rules per Security Group	1000
IPs or Subnets per Security Group Rule	4000
Security Groups per Port/Server	28

Creating Security Groups

Create security groups to manage network traffic for one or more instances.

Prerequisites

- You have the `network admin` role.

  For more information, see Authorizations.

- You know and comply with the configuration limits for security groups.

  For more information, see Security Groups Configuration Maximum.

Procedure

1. Choose **Networking & Loadbalancing** > **Security Groups**.
2. Choose **Create New**.

   A dialog box opens.
3. Enter a name.
4. Enter a description.
5. Choose **Create**.

Results

The security group is created and automatically configured to allow all outgoing (egress) connections.

Next Steps

Adding Security Group Rules to Security Groups

Adding Security Group Rules to Security Groups

Security group rules specify the network traffic which is allowed for all instances within a security group.

Prerequisites

- You have the `network admin` role.

  For more information, see Authorizations.

- You know and comply with the configuration limits for security groups.

  For more information, see Security Groups Configuration Maximum.

- You know the network traffic rules.

  For more information, see Network Traffic Rules.

Context

A common use case for adding security group rules is to allow remote access to server instances by creating an ingress rule for SSH (port 22) or RDP (port 3389).

Procedure

1. Choose Networking & Loadbalancing > Security Groups.
2. Choose the security group.
3. Choose **Add New Rule**.
4. Select the rule type.
5. Select the protocol.
6. Select the direction. You have the following options:
   - **Ingress**

     An ingress rule defines which origin networks are allowed to connect to the instance and on which ports.

   - **Egress**

     An egress rule defines which target networks the instance is allowed to connect to.

7. Enter a specific port, such as 80, or a port range, such as 0-80.

   SAP Cloud Infrastructure enforces port security for all supported ports.
8. Select the remote source of the network traffic. You have the following options:
   - **IP Address**
   - **Security Group**
9. Enter the IP address, IP range, or security group.

   ### Example

   - Specific IP address: `10.180.1.1/32`
   - IP range: `10.180.1.0/24`

     #### Caution

     `0.0.0.0/0` allows network traffic from or to any IP address.

10. Choose **Create**.

Sharing Security Groups with other Projects

Define which target projects can use a security group. This enables reuse across multiple projects without duplicating settings and prevents target projects from modifying them.

Prerequisites

- You have the `network admin` role.

  For more information, see Authorizations.

- You have a security group other than the **default** security group.

  Note

  You cannot share the **default** security group.

  For more information, see Creating Security Groups.

Procedure

1. Choose **Networking & Loadbalancing** > **Security Groups**.
2. From the gear dropdown menu for the security group, choose **Access Control**.

   A dialog box opens.
3. Enter the project name or ID of the target project.
4. Choose **Add**.

Deleting Security Group Rules from Security Groups

Delete security rules that you no longer need from a security group.

Prerequisites

- You have the `network admin` role.

  For more information, see Authorizations.

Procedure

1. Choose **Networking & Loadbalancing** > **Security Groups**.
2. Choose the security group.
3. Choose the delete button next to the security rule you want to delete.
4. Confirm the dialog box.

Deleting Security Groups

Permanently delete a security group you no longer need.

Prerequisites

- You have the `network admin` role.

  For more information, see [Authorizations](#).

Context

Ensure that you delete any unused security groups that aren't attached to any instances. Deleting unused security groups not only keeps your environment clean but also ensures that you don't accidentally attach them to any instances, which could inadvertently open up your environment to attacks.

Note

You can't delete the **default** security group.

Procedure

1. Choose **Networking & Loadbalancing** > **Security Groups**.
2. From the gear dropdown menu for the security group, choose **Delete**.
3. Confirm the dialog box.

Load Balancers

Load balancers distribute incoming network traffic across multiple server instances to improve availability and prevent overload. Configure load balancers to manage traffic flow, monitor health, and ensure efficient resource use.

[Load Balancer Statuses](#)

Get an overview of the possible statuses for load balancers, listeners, pools, pool members, and health monitors.

[Creating Load Balancers](#)

Create load balancers to automatically distribute network traffic across multiple server instances.

[Editing Load Balancers](#)

Update the name, description, and tags of an existing load balancer.

[Deleting Load Balancers](#)

Permanently delete a load balancer you no longer need.

[Viewing Load Balancer JSON Configuration](#)

View the JSON configuration of the load balancer.

[Adding Floating IPs to Load Balancers](#)

Add a floating IP to a load balancer.

[Removing Floating IPs from Load Balancers](#)

Remove a floating IP that is no longer needed from a load balancer.

[Listeners](#)

Listeners define how the load balancer processes incoming traffic by specifying protocols and ports. Create and configure listeners, and apply traffic management policies to control how traffic is directed to backend resources.

[Pools](#)

Pools define backend server groups for traffic distribution. Configure and manage pool members, monitor health, and adjust settings.

Load Balancer Statuses

Get an overview of the possible statuses for load balancers, listeners, pools, pool members, and health monitors.

Note

Status updates are subject to the polling interval. After changing load balancer objects, allow up to 2 minutes for the status to update.

Load Balancer Statuses

Load Balancer Status	Description	API Status Equivalent
ONLINE	The load balancer is operational and serving traffic.	ACTIVE
OFFLINE	The load balancer is not serving traffic, possibly due to unhealthy amphorae, errors, or all pool members being unhealthy.	ERROR, non-operational state
PENDING	The load balancer is being created, updated, or deleted.	PENDING_CREATE, PENDING_UPDATE, PENDING_DELETE
DEGRADED	Some objects in the hierarchy (such as pool members) are unhealthy (for example, OFFLINE), but the load balancer is still operational.	ACTIVE (with unhealthy components)

Listener Statuses

Listener Status	Description	API Status Equivalent
ONLINE	The listener is configured and ready to accept traffic, but requires an operational load balancer.	ACTIVE
OFFLINE	The listener is not operational, possibly due to errors or being disabled.	ERROR, non-operational state
PENDING	The listener is being created, updated, or deleted.	PENDING_CREATE, PENDING_UPDATE, PENDING_DELETE
DEGRADED	The listener has reduced functionality, typically during failover in Active/Standby topology (context-specific).	

Pool Statuses

Pool Status	Description	API Status Equivalent
ONLINE	The pool is operational with at least some healthy members.	ACTIVE
OFFLINE	The pool is not operational, often due to all members being unhealthy or errors.	ERROR
DEGRADED	Some members are unhealthy (for example, OFFLINE), but the pool is still operational.	ACTIVE (with unhealthy members)
PENDING	The pool is being created, updated, or deleted.	PENDING_CREATE, PENDING_UPDATE, PENDING_DELETE

Pool Member Statuses

Pool Member Status	Description	API Status Equivalent
ONLINE	The member is healthy and receiving traffic.	ONLINE
OFFLINE	The member is not receiving traffic due to configuration or health check failure.	OFFLINE
ERROR	The member has failed health checks.	ERROR
DRAINING	The member is being gracefully removed, allowing existing connections to complete.	DRAINING
NO_MONITOR	No health monitor is configured, so health is unknown.	NO_MONITOR

Health Monitor Statuses

Health Monitor Status	Description	API Status Equivalent
ONLINE	The health monitor is actively checking member health.	ONLINE
OFFLINE	The health monitor is disabled or not running.	OFFLINE
ERROR	An error occurred during monitoring.	ERROR

Creating Load Balancers

Create load balancers to automatically distribute network traffic across multiple server instances.

Prerequisites

- You have the `loadbalancer_admin` role.

  For more information, see Authorizations.

- You have sufficient quota. To create a load balancer, you need quota for one to three ports: one for the virtual IP (VIP) and two for SelfIP addresses. The two SelfIP addresses are only required if the load balancer is the first one in its subnet.

  For more information, see Resource Management.

Procedure

1. Choose Networking & Loadbalancing > Load Balancers.
2. Choose **New Load Balancer**.

   A dialog box opens.
3. Enter a name.
4. **Optional:** Enter a description.
5. Select a private network.
6. **Optional:** To configure advanced network options, choose **Toggle advanced network options**.
   a. Select the subnet for the load balancer.
   b. To automatically allocate a new IP address from the private network, leave the IP address field empty, or enter a specific IP address.
7. **Optional:** Enter tags to organize or classify the load balancer.
8. Choose **Save**.

Results

- The activity is recorded as an event in the audit log.

Next Steps

- Listeners
- Pools

Related Information

Audit

Editing Load Balancers

Update the name, description, and tags of an existing load balancer.

Prerequisites

- You have the `loadbalancer_admin` role.

  For more information, see Authorizations.

Procedure

1. Choose Networking & Loadbalancing > Load Balancers.
2. From the gear dropdown menu for the load balancer, choose **Edit**.

A dialog box opens.

3. Enter a name.
4. **Optional:** Enter a description.
5. **Optional:** Enter tags to organize or classify the load balancer.
6. Choose **Save**.

Results

- The activity is recorded as an event in the audit log.

Related Information

[Audit]

Deleting Load Balancers

Permanently delete a load balancer you no longer need.

Prerequisites

- You have the `loadbalancer_admin` role.

  For more information, see [Authorizations].

Procedure

1. Choose **Networking & Loadbalancing** > **Load Balancers**.
2. From the gear dropdown menu for the load balancer, choose **Delete**.
3. Confirm the dialog box.

Results

- The activity is recorded as an event in the audit log.

Related Information

[Audit]

Viewing Load Balancer JSON Configuration

View the JSON configuration of the load balancer.

Prerequisites

- You have the `loadbalancer_viewer` role.

  For more information, see [Authorizations].

Procedure

1. Choose Networking & Loadbalancing > Load Balancers.
2. From the gear dropdown menu for the load balancer, choose **JSON**.

   A dialog box opens and shows the load balancer JSON configuration.

Adding Floating IPs to Load Balancers

Add a floating IP to a load balancer.

Prerequisites

- You have the `loadbalancer_admin` role.

  For more information, see Authorizations.

- You have a floating IP.

  For more information, see Floating IPs.

Procedure

1. Choose Networking & Loadbalancing > Load Balancers.
2. From the gear dropdown menu for the load balancer, choose **Attach Floating IP**.

   A dialog box opens.
3. Select the floating IP.
4. Choose **Save**.

Results

- The activity is recorded as an event in the audit log.

Related Information

Audit

Removing Floating IPs from Load Balancers

Remove a floating IP that is no longer needed from a load balancer.

Prerequisites

- You have the `loadbalancer_admin` role.

  For more information, see Authorizations.

Procedure

1. Choose Networking & Loadbalancing > Load Balancers.

2. From the gear dropdown menu for the load balancer, choose **Detach Floating IP**.

Results

- The activity is recorded as an event in the audit log.

Related Information

Audit

Listeners

Listeners define how the load balancer processes incoming traffic by specifying protocols and ports. Create and configure listeners, and apply traffic management policies to control how traffic is directed to backend resources.

Extended Policies

Get an overview of the extended policies available for your load balancer. These policies allow you to fine-tune how traffic is managed and processed, from enabling compression, encryption, and proxy protocols to controlling header overrides and protocol settings.

TLS Cipher Suite Catalog

Get an overview of the default cipher suites attached to listeners using TLS-terminated HTTPS protocol. Each cipher suite defines specific encryption algorithms and protocols used for securing network traffic.

HTTP and TLS-Terminated HTTPS Listener Headers

Get an overview of the optional headers you can insert into HTTP and TLS-terminated HTTPS listeners. These headers provide additional information about the incoming requests, which can be forwarded to the pool members.

Creating HTTP Listeners

Add a listener for the hypertext transfer protocol (HTTP) to forward HTTP-based traffic to the backend pool.

Creating TCP Listeners

Add a listener for the Transmission Control Protocol (TCP) to forward raw TCP traffic to a backend pool.

Creating TLS-Terminated HTTPS Listeners

Add a listener for the TLS-terminated hypertext transfer protocol (HTTPS) to handle encrypted HTTPS traffic, decrypt it at the load balancer, and forward it to the backend pool.

Creating UDP Listeners

Add a User Datagram Protocol (UDP) listener to forward UDP-based traffic (for example, DNS) to the backend pool.

Editing Listeners

Update the name, description, default pool, or connection limit of an existing listener.

Deleting Listeners

Delete listeners that you no longer need.

Configuring Allowed CIDRs for Listeners

Restrict traffic by filtering source IP addresses and subnets.

Viewing Listener JSON Configuration

View the JSON configuration of the listener.

Creating L7 Policies

Create L7 policies to block or reroute traffic based on L7 rules.

Editing L7 Policies

Update the name, description, position, action, or tags of an existing L7 policy.

Deleting L7 Policies

Delete L7 policies that you no longer need.

Viewing L7 Policy JSON Configuration

View the JSON configuration of the L7 policy.

[Creating L7 Rules]

Create L7 rules to manage how the load balancer handles application-layer traffic, such as HTTP requests, based on specific request characteristics. These rules are part of an L7 policy, and all rules in a policy are logically ANDed together.

[Editing L7 Rules]

Update the rule type, comparison type, value, and tags of an existing L7 rule.

[Deleting L7 Rules]

Delete L7 rules that you no longer need.

[Viewing L7 Rule JSON Configuration]

View the JSON configuration of the L7 rule.

Extended Policies

Get an overview of the extended policies available for your load balancer. These policies allow you to fine-tune how traffic is managed and processed, from enabling compression, encryption, and proxy protocols to controlling header overrides and protocol settings.

Extended Policy	Tag	Protocol Type	Description
Enable HTTP Compression	http_compression_e4a2_v1_0	HTTP TERMINATED_HTTPS	Apply a compression profile to your listener to compress all content of type text/* and application/(xml|x-javascript) using the gzip method. Make sure your content is at least 1024 bytes long to enable compression.
Enable Cookie Encryption	cookie_encryption_b82a_v1_0	HTTP TERMINATED_HTTPS	Encrypt all cookies that are sent to your client and automatically decrypt them for the pool members.
Enable Proxy Protocol v1	proxy_protocol_2edF_v1_0	TCP HTTPS HTTP TERMINATED_HTTPS	Enable version 1 of the proxy protocol for your listener. Ensure your pool members support this protocol to avoid backend errors. You can also apply this setting to listeners with protocol TCP if you have the tag standard_tcp_a3de_v1_0 set.
Enable Proxy Protocol v2	proxy_protocol_V2_e8f6_v1_0	TCP HTTPS HTTP TERMINATED_HTTPS	Enable version 2 of the proxy protocol for your listener. Make sure your pool members support this protocol to prevent errors on the backend. You can also apply this to listeners with protocol TCP if the tag standard_tcp_a3de_v1_0 is set.
Disable HW Acceleration	standard_tcp_a3de_v1_0	TCP	Disable hardware acceleration for listeners using the TCP or

Extended Policy	Tag	Protocol Type	Description
		HTTPS	HTTPS protocol. Disabling is required, for example, when you need to use the proxy protocol (either version).
Enable HTTP to HTTPS Redirect	http_redirect_a26c_v1_0	HTTP	Automatically redirect all HTTP traffic to HTTPS to secure your connections.
Enable Stateless UDP Load Balancing	ccloud_special_udp_stateless	UDP	By default, your load balancer sends every UDP packet from a client to the same pool member, which is helpful for longer client-server communications. Enable stateless UDP load balancing to distribute UDP packets across multiple pool members and improve scalability and performance.
Disable SNAT	ccloud_special_l4_deactivate_snat	TCP HTTPS HTTP TERMINATED_HTTPS UDP	By default, your load balancer uses source network address translation (SNAT). Use this setting to disable SNAT, so your pool member can see the original IP address of the client. **Note** When you disable SNAT, make sure your pool members have routes that return traffic to the load balancer IP address. Without this, TCP connections will fail, as the initiator expects ACK packets from the load balancer IP, not directly from the pool member.
Override X-Forwarded-Host Header	ccloud_special_xfh_override	HTTP TERMINATED_HTTPS	Ensure full control over the X-Forwarded-Host header. If this header is present in an incoming HTTP(S) request, the load balancer overrides its value with the value from the built-in HTTP Host header. This helps you prevent malicious header injection by clients.
Enable TCP Connection Mirroring	ccloud_special_tcp_mirror	TCP	Enable mirroring of TCP connections between active and standby devices in the load balancer infrastructure. This helps maintain your TCP connections without

Extended Policy	Tag	Protocol Type	Description
			interruption during maintenance, failovers, or reboots. Use this option if you need high resiliency for long-lived TCP connections. **Note** TCP connection mirroring is only supported for listeners with the TCP protocol and is not suitable for short-lived connections like typical HTTP(S) traffic.

Related Information

Creating HTTP Listeners
Creating TCP Listeners
Creating TLS-Terminated HTTPS Listeners
Creating UDP Listeners

TLS Cipher Suite Catalog

Get an overview of the default cipher suites attached to listeners using TLS-terminated HTTPS protocol. Each cipher suite defines specific encryption algorithms and protocols used for securing network traffic.

Cipher Suite	Cipher Type	Description
AES128-GCM-SHA256	AES	AES128 encryption in GCM mode with SHA-256 for message authentication
AES128-SHA	AES	AES128 encryption with SHA for message authentication
AES128-SHA256	AES	AES128 encryption with SHA-256 for message authentication
AES256-GCM-SHA384	AES	AES256 encryption in GCM mode with SHA-384 for message authentication
AES256-SHA	AES	AES256 encryption with SHA for message authentication
AES256-SHA256	AES	AES256 encryption with SHA-256 for message authentication
CAMELLIA128-SHA	Camellia	Camellia128 encryption with SHA for message authentication
CAMELLIA256-SHA	Camellia	Camellia256 encryption with SHA for message authentication
ECDHE-ECDSA-AES128-GCM-SHA256	ECDHE-ECDSA	ECDHE key exchange with ECDSA authentication, AES128 encryption in GCM

Cipher Suite	Cipher Type	Description
		mode, and SHA-256 for message authentication
ECDHE-ECDSA-AES128-SHA	ECDHE-ECDSA	ECDHE key exchange with ECDSA authentication, AES128 encryption, and SHA for message authentication
ECDHE-ECDSA-AES128-SHA256	ECDHE-ECDSA	ECDHE key exchange with ECDSA authentication, AES128 encryption, and SHA-256 for message authentication
ECDHE-ECDSA-AES256-GCM-SHA384	ECDHE-ECDSA	ECDHE key exchange with ECDSA authentication, AES256 encryption in GCM mode, and SHA-384 for message authentication
ECDHE-ECDSA-AES256-SHA	ECDHE-ECDSA	ECDHE key exchange with ECDSA authentication, AES256 encryption, and SHA for message authentication
ECDHE-ECDSA-AES256-SHA384	ECDHE-ECDSA	ECDHE key exchange with ECDSA authentication, AES256 encryption, and SHA-384 for message authentication
ECDHE-RSA-AES128-CBC-SHA	ECDHE-RSA	ECDHE key exchange with RSA authentication, AES128 encryption in CBC mode, and SHA for message authentication
ECDHE-RSA-AES128-GCM-SHA256	ECDHE-ECDSA	ECDHE key exchange with RSA authentication, AES128 encryption in GCM mode, and SHA-256 for message authentication
ECDHE-RSA-AES128-SHA256	ECDHE-RSA	ECDHE key exchange with RSA authentication, AES128 encryption, and SHA-256 for message authentication
ECDHE-RSA-AES256-CBC-SHA	ECDHE-RSA	ECDHE key exchange with RSA authentication, AES256 encryption in CBC mode, and SHA for message authentication
ECDHE-RSA-AES256-GCM-SHA384	ECDHE-RSA	ECDHE key exchange with RSA authentication, AES256 encryption in GCM mode, and SHA-384 for message authentication
ECDHE-RSA-AES256-SHA384	ECDHE-RSA	ECDHE key exchange with RSA authentication, AES256 encryption, and SHA-384 for message authentication
ECDHE-RSA-CHACHA20-POLY1305-SHA256	ECDHE-RSA	ECDHE key exchange with RSA authentication, ChaCha20 encryption, and Poly1305 for message authentication
TLS13-AES128-GCM-SHA256	TLS 1.3	TLS 1.3 cipher suite with AES128 encryption in GCM mode and SHA-256 for message authentication

Cipher Suite	Cipher Type	Description
TLS13-AES256-GCM-SHA384	TLS 1.3	TLS 1.3 cipher suite with AES256 encryption in GCM mode and SHA-384 for message authentication
TLS13-CHACHA20-POLY1305-SHA256	TLS 1.3	TLS 1.3 cipher suite with ChaCha20 encryption and Poly1305 for message authentication

HTTP and TLS-Terminated HTTPS Listener Headers

Get an overview of the optional headers you can insert into HTTP and TLS-terminated HTTPS listeners. These headers provide additional information about the incoming requests, which can be forwarded to the pool members.

Header	Description	Protocol Type
X-Forwarded-For	Adds the original client IP address.	HTTP TERMINATED_HTTPS
X-Forwarded-Port	Adds the original port number used by the client.	HTTP TERMINATED_HTTPS
X-Forwarded-Proto	Adds the original protocol (HTTP or HTTPS).	HTTP TERMINATED_HTTPS
X-SSL-Client-Verify	Indicates the verification status of the client certificate.	TERMINATED_HTTPS
X-SSL-Has-Cert	Indicates whether the client certificate is present.	TERMINATED_HTTPS
X-SSL-Client-DN	Adds the distinguished name (DN) of the client certificate.	TERMINATED_HTTPS
X-SSL-Client-CN	Adds the common name (CN) of the client certificate.	TERMINATED_HTTPS
X-SSL-Issuer	Adds the issuer of the client certificate.	TERMINATED_HTTPS
X-SSL-ClientSHA1	Adds the SHA1 fingerprint of the client certificate.	TERMINATED_HTTPS
X-SSL-Client-Not-Before	Adds the "Not Before" timestamp from the client certificate.	TERMINATED_HTTPS
X-SSL-Client-Not-After	Adds the "Not After" timestamp from the client certificate.	TERMINATED_HTTPS

Creating HTTP Listeners

Add a listener for the hypertext transfer protocol (HTTP) to forward HTTP-based traffic to the backend pool.

Prerequisites

- You have the `loadbalancer_admin` role.

  For more information, see Authorizations.

- You have a load balancer.

- **Optional:** If you want to apply an extended policy, make sure you are familiar with the available extended policies and understand their impact.

  For more information, see Extended Policies.

- **Optional:** If you want to insert headers into the request before it is sent to the pool member, ensure you are familiar with the available headers and their meaning.

  For more information, see HTTP and TLS-Terminated HTTPS Listener Headers.

Procedure

1. Choose **Networking & Loadbalancing** > **Load Balancers**.
2. Choose the load balancer to which you want to add a listener.
3. Choose **New Listener**.

   A dialog box opens.
4. Enter a name.
5. **Optional:** Enter a description.
6. Enter a protocol port number between 1 and 65535 to reach the load balancer.
7. Select the protocol: **HTTP**
8. **Optional:** To apply predefined policies for special purposes, select an extended policy.

   The extended policy will always apply, and L7 rules are not applicable.

   The extended policy is displayed as a tag.
9. **Optional:** Insert headers into the request before it is sent to the pool member.
10. **Optional:** Select a default pool.

    All traffic routes to the default pool if no L7 policy defines a different pool. If you do not yet have any pools, leave the field empty.
11. **Optional:** Adjust the connection limit.
12. **Optional:** Enter tags to organize or classify listeners, or to add extended policies.
13. Choose **Save**.

Results

- The activity is recorded as an event in the audit log.

Next Steps

Creating L7 Policies

Related Information

Audit

Creating TCP Listeners

Add a listener for the Transmission Control Protocol (TCP) to forward raw TCP traffic to a backend pool.

Prerequisites

- You have the `loadbalancer_admin` role.

  For more information, see Authorizations.

- You have a load balancer.

- **Optional:** If you want to apply an extended policy, make sure you are familiar with the available extended policies and understand their impact.

  For more information, see Extended Policies.

Procedure

1. Choose **Networking & Loadbalancing** > **Load Balancers**.
2. Choose the load balancer to which you want to add a listener.
3. Choose **New Listener**.

   A dialog box opens.
4. Enter a name.
5. **Optional:** Enter a description.
6. Enter a protocol port number between 1 and 65535 to reach the load balancer.
7. Select the protocol: **TCP**
8. **Optional:** To apply predefined policies for special purposes, select an extended policy.

   The extended policy will always apply, and L7 rules are not applicable.

   The extended policy is displayed as a tag.
9. **Optional:** Select a default pool.

   All traffic routes to the default pool if no L7 policy defines a different pool. If you do not yet have any pools, leave the field empty.
10. **Optional:** Adjust the connection limit.
11. **Optional:** Enter tags to organize or classify listeners, or to add extended policies.
12. Choose **Save**.

Results

- The activity is recorded as an event in the audit log.

Related Information

Audit

Creating TLS-Terminated HTTPS Listeners

Add a listener for the TLS-terminated hypertext transfer protocol (HTTPS) to handle encrypted HTTPS traffic, decrypt it at the load balancer, and forward it to the backend pool.

Prerequisites

- You have the `loadbalancer_admin` role.

  For more information, see Authorizations.

- You have a load balancer.

- You have a secret containing a PKCS12 format certificate/key.

  For more information, see Key Manager.

- **Optional:** If you want to apply an extended policy, make sure you are familiar with the available extended policies and understand their impact.

  For more information, see Extended Policies.

- **Optional:** If you want to insert headers into the request before it is sent to the pool member, ensure you are familiar with the available headers and their meaning.

  For more information, see HTTP and TLS-Terminated HTTPS Listener Headers.

- **Optional:** If you want to change the default cipher suites attached to listeners, make sure you are familiar with the cipher suites and understand their implications.

  For more information, see TLS Cipher Suite Catalog.

Procedure

1. Choose Networking & Loadbalancing > Load Balancers.
2. Choose the load balancer to which you want to add a listener.
3. Choose New Listener.

   A dialog box opens.
4. Enter a name.
5. **Optional:** Enter a description.
6. Enter a protocol port number between 1 and 65535 to reach the load balancer.
7. Select the protocol: TERMINATED_HTTPS
8. Select the secret containing a PKCS12 format certificate/key.
9. **Optional:** To apply predefined policies for special purposes, select an extended policy.

   The extended policy is displayed as a tag.
10. **Optional:** Insert headers into the request before it is sent to the pool member.
11. **Optional:** When you have multiple TLS certificates that you would like to use on the same listener, select the server name indication (SNI) secrets.
12. **Optional:** Select the client authentication mode. You have the following options:
    - NONE
    - OPTIONAL
    - MANDATORY

13. **Optional:** Select the client authentication secret.
14. **Optional:** Remove default TLS ciphers suites attached to the listener.
15. **Optional:** Select a default pool.

    All traffic routes to the default pool if no L7 policy defines a different pool. If you do not yet have any pools, leave the field empty.
16. **Optional:** Adjust the connection limit.
17. **Optional:** Enter tags to organize or classify listeners, or to add extended policies.
18. Choose **Save**.

Results

- The activity is recorded as an event in the audit log.

Next Steps

Creating L7 Policies

Related Information

Audit

Creating UDP Listeners

Add a User Datagram Protocol (UDP) listener to forward UDP-based traffic (for example, DNS) to the backend pool.

Prerequisites

- You have the `loadbalancer_admin` role.

  For more information, see Authorizations.
- You have a load balancer.
- **Optional:** If you want to apply an extended policy, make sure you are familiar with the available extended policies and understand their impact.

  For more information, see Extended Policies.

Procedure

1. Choose **Networking & Loadbalancing** > **Load Balancers**.
2. Choose the load balancer to which you want to add a listener.
3. Choose **New Listener**.

   A dialog box opens.
4. Enter a name.
5. **Optional:** Enter a description.
6. Enter a protocol port number between 1 and 65535 to reach the load balancer.
7. Select the protocol: **UDP**

8. **Optional:** Select a default pool.

   All traffic routes to the default pool if no L7 policy defines a different pool. If you do not yet have any pools, leave the field empty.

9. **Optional:** Adjust the connection limit.

10. **Optional:** Enter tags to organize or classify listeners, or to add extended policies.

11. Choose **Save**.

Results

- The activity is recorded as an event in the audit log.

Related Information

Audit

Editing Listeners

Update the name, description, default pool, or connection limit of an existing listener.

Prerequisites

- You have the `loadbalancer_admin` role.

  For more information, see Authorizations.

Procedure

1. Choose **Networking & Loadbalancing** > **Load Balancers**.

2. Choose the load balancer that has listeners.

3. From the gear dropdown menu for the listener, choose **Edit**.

   A dialog box opens.

4. Update any fields.

5. Choose **Save**.

Results

- The activity is recorded as an event in the audit log.

Related Information

Audit

Deleting Listeners

Delete listeners that you no longer need.

Prerequisites

- You have the `loadbalancer_admin` role.

  For more information, see [Authorizations].

Procedure

1. Choose **Networking & Loadbalancing** > **Load Balancers**.
2. Choose the load balancer that has listeners.
3. From the gear dropdown menu for the listener, choose **Delete**.

Results

- The activity is recorded as an event in the audit log.

Related Information

Audit

Configuring Allowed CIDRs for Listeners

Restrict traffic by filtering source IP addresses and subnets.

Prerequisites

- You have the `loadbalancer_admin` role.

  For more information, see Authorizations.

Context

By default, all traffic is allowed. If you specify a list of CIDRs, the default behavior changes to deny all traffic except for the specified CIDRs.

Procedure

1. Choose **Networking & Loadbalancing** > **Load Balancers**.
2. Choose the load balancer that contains the listener you want to configure.
3. From the gear dropdown menu for the listener, choose **Allowed CIDRs**.

   A dialog box opens.
4. Enter one or more IPv4 or IPv6 CIDRs.
5. Choose **Save**.

Results

- Once a list of CIDRs is provided, all other traffic is denied by default.

- The activity is recorded as an event in the audit log.

Related Information

[Audit]

Viewing Listener JSON Configuration

View the JSON configuration of the listener.

Prerequisites

- You have the `loadbalancer_viewer` role.

  For more information, see [Authorizations].

Procedure

1. Choose **Networking & Loadbalancing** > **Load Balancers**.
2. Choose the load balancer that has listeners.
3. From the gear dropdown menu for the listener, choose **JSON**.

   A dialog box opens and shows the listener JSON configuration.

Creating L7 Policies

Create L7 policies to block or reroute traffic based on L7 rules.

Prerequisites

- You have the `loadbalancer_admin` role.

  For more information, see [Authorizations].

- You have a load balancer.

  For more information, see [Creating Load Balancers].

- You have an HTTP or TLS-terminated HTTPS listener.

  For more information, see:

  - [Creating HTTP Listeners]
  - [Creating TLS-Terminated HTTPS Listeners]

Procedure

1. Choose **Networking & Loadbalancing** > **Load Balancers**.
2. Choose the load balancer containing the listener to which you want to add a L7 policy.
3. Choose the listener.
4. Choose **New L7 Policy**.

   A dialog box opens.

5. Enter a name.
6. **Optional:** Enter a description.
7. **Optional:** Change the position of the policy.

   L7 policies are evaluated in the order defined by the position attribute. The first policy that matches a request determines the action taken. If no policy matches, the request routes to the listener's default pool, if it exists.
8. Select an action that will be executed when all L7 policies are matched. You have the following options:

Action	Description
REDIRECT_PREFIX	Redirect requests to a specified prefix. Select the redirect HTTP code (301, 302, 303, 307, or 308) and enter the redirect prefix.
REDIRECT_TO_POOL	Redirect requests to a specific pool. Enter the pool ID.
REDIRECT_TO_URL	Redirect requests to a specific URL. Select the redirect HTTP code (301, 302, 303, 307, or 308) and the URL.
REJECT	Reject the request.

9. **Optional:** Enter tags to organize or classify the L7 policy.
10. Choose **Save**.

Results

- The activity is recorded as an event in the audit log.

Next Steps

- [Creating L7 Rules]

Related Information

[Audit]

Editing L7 Policies

Update the name, description, position, action, or tags of an existing L7 policy.

Prerequisites

- You have the `loadbalancer_admin` role.

  For more information, see Authorizations.

Procedure

1. Choose Networking & Loadbalancing > Load Balancers.
2. Choose the load balancer.
3. Choose the listener.
4. From the gear dropdown menu for the policy, choose **Edit**.

   A dialog box opens.
5. Enter a name.
6. **Optional:** Enter a description.
7. **Optional:** Change the position of the policy.

   L7 policies are evaluated in the order defined by the position attribute. The first policy that matches a request determines the action taken. If no policy matches, the request routes to the listener's default pool, if it exists.
8. Select an action that will be executed when all L7 policies are matched. You have the following options:

Action	Description
REDIRECT_PREFIX	Redirect requests to a specified prefix. Select the redirect HTTP code (301, 302, 303, 307, or 308) and enter the redirect prefix.
REDIRECT_TO_POOL	Redirect requests to a specific pool. Enter the pool ID.
REDIRECT_TO_URL	Redirect requests to a specific URL. Select the redirect HTTP code (301, 302, 303, 307, or 308) and the URL.
REJECT	Reject the request.

9. **Optional:** Enter tags to organize or classify the L7 policy.
10. Choose **Save**.

Results

- The activity is recorded as an event in the audit log.

Related Information

Audit

Deleting L7 Policies

Delete L7 policies that you no longer need.

Prerequisites

- You have the `loadbalancer_admin` role.

  For more information, see Authorizations.

- You have deleted all L7 rules from the L7 policy.

  For more information, see Deleting L7 Rules.

Procedure

1. Choose **Networking & Loadbalancing** > **Load Balancers**.
2. Choose the load balancer containing the listener from which you want to delete a L7 policy.
3. Choose the listener.
4. From the gear dropdown menu for the L7 policy, choose **Delete**.

Results

- The activity is recorded as an event in the audit log.

Related Information

Audit

Viewing L7 Policy JSON Configuration

View the JSON configuration of the L7 policy.

Prerequisites

- You have the `loadbalancer_viewer` role.

  For more information, see Authorizations.

Procedure

1. Choose **Networking & Loadbalancing** > **Load Balancers**.
2. Choose the load balancer.
3. Choose the listener.
4. From the gear dropdown menu for the L7 policy, choose **JSON**.

   A dialog box opens and shows the L7 policy JSON configuration.

Creating L7 Rules

Create L7 rules to manage how the load balancer handles application-layer traffic, such as HTTP requests, based on specific request characteristics. These rules are part of an L7 policy, and all rules in a policy are logically ANDed together.

Prerequisites

- You have the `loadbalancer_admin` role.

  For more information, see Authorizations.

- You have an L7 policy.

  For more information, see Creating L7 Policies.

Context

Without L7 rules, requests route to the listener's default pool. L7 rules let the listener evaluate requests and make routing decisions based on specific criteria. This eliminates the need for multiple load balancers.

Procedure

1. Choose Networking & Loadbalancing > Load Balancers.
2. Choose the load balancer.
3. Choose the listener.
4. Choose the policy.
5. Choose New L7 Rule.

   A dialog box opens.
6. Select the rule type. You have the following options:

Rule Type	Description
COOKIE	Matches against a cookie name or value.
FILE_TYPE	Matches based on the file extension in the request URI.
HEADER	Matches the value of a specific HTTP header.
HOST_NAME	Matches the hostname from the HTTP request.
PATH	Matches the request URI path.
SSL_CONN_HAS_CERT	Matches whether the SSL connection presents a client certificate.
SSL_VERIFY_RESULT	Matches the result of the SSL certificate verification.
SSL_DN_FIELD	Matches fields in the SSL certificate's Distinguished Name (DN), such as CN or O.

7. Select the compare type. You have the following options:

Compare Type	Description
CONTAINS	Checks if the request field includes the specified value.
ENDS_WITH	Checks if the request field ends with the specified value.
EQUAL_TO	Checks if the request field exactly matches the specified value.
STARTS_WITH	Checks if the request field starts with the specified value.

8. **Optional:** To invert the logic of the rule, select the Inverse Comparisation (NOT) checkbox.

Example

If you select the comparison type EQUAL_TO and select inverse comparison, the rule matches when the values are not equal.

9. Enter the value to compare against, based on the selected rule type.
10. **Optional:** Enter tags to organize or classify the rule.
11. Choose Save.

Results

- The activity is recorded as an event in the audit log.

Related Information

Audit

Editing L7 Rules

Update the rule type, comparison type, value, and tags of an existing L7 rule.

Prerequisites

- You have the `loadbalancer_admin` role.

  For more information, see Authorizations.

Procedure

1. Choose Networking & Loadbalancing > Load Balancers.
2. Choose the load balancer.
3. Choose the listener.
4. Choose the policy.
5. From the gear dropdown menu for the rule, choose Edit.

   A dialog box opens.
6. Select the rule type. You have the following options:

Rule Type	Description
COOKIE	Matches against a cookie name or value.
FILE_TYPE	Matches based on the file extension in the request URI.
HEADER	Matches the value of a specific HTTP header.
HOST_NAME	Matches the hostname from the HTTP request.
PATH	Matches the request URI path.
SSL_CONN_HAS_CERT	Matches whether the SSL connection presents a client certificate.

Rule Type	Description
SSL_VERIFY_RESULT	Matches the result of the SSL certificate verification.
SSL_DN_FIELD	Matches fields in the SSL certificate's Distinguished Name (DN), such as CN or O.

7. Select the compare type. You have the following options:

Compare Type	Description
CONTAINS	Checks if the request field includes the specified value.
ENDS_WITH	Checks if the request field ends with the specified value.
EQUAL_TO	Checks if the request field exactly matches the specified value.
STARTS_WITH	Checks if the request field starts with the specified value.

8. **Optional:** To invert the logic of the rule, select the Inverse Comparisation (NOT) checkbox.

   Example

   If you select the comparison type EQUAL_TO and select inverse comparison, the rule matches when the values are not equal.

9. Enter the value to compare against, based on the selected rule type.
10. **Optional:** Enter tags to organize or classify the rule.
11. Choose Save.

Results

- The activity is recorded as an event in the audit log.

Related Information

Audit

Deleting L7 Rules

Delete L7 rules that you no longer need.

Prerequisites

- You have the `loadbalancer_admin` role.

  For more information, see Authorizations.

Procedure

1. Choose Networking & Loadbalancing > Load Balancers.
2. Choose the load balancer.

3. Choose the listener.
4. Choose the policy.
5. From the gear dropdown menu for the L7 rule, choose **Delete**.
6. Confirm the dialog box.

Results

- The activity is recorded as an event in the audit log.

Related Information

Audit

Viewing L7 Rule JSON Configuration

View the JSON configuration of the L7 rule.

Prerequisites

- You have the `loadbalancer_viewer` role.

  For more information, see Authorizations.

Procedure

1. Choose **Networking & Loadbalancing** > **Load Balancers**.
2. Choose the load balancer.
3. Choose the listener.
4. Choose the policy.
5. From the gear dropdown menu for the L7 rule, choose **JSON**.

   A dialog box opens and shows the L7 rule JSON configuration.

Pools

Pools define backend server groups for traffic distribution. Configure and manage pool members, monitor health, and adjust settings.

Creating Pools

Create pools to define which pool members (your backend servers) the load balancer sends traffic to. You also select how the load balancer distributes traffic between these pool members.

Editing Pools

Update the name, description, algorithm, session persistence type, or tags of an existing pool.

Deleting Pools

Permanently delete a pool you no longer need.

Viewing Pool JSON Configuration

View the JSON configuration of the pool.

[Creating Pool Members](#)

Create a new pool member that serves the traffic behind a load balancer. Each pool member is indicated by the IP address and port that it uses for traffic.

[Editing Pool Members](#)

Update the name, IPs, weight, or tags of an existing pool member.

[Deleting Pool Members](#)

Delete pool members that you no longer need.

[Viewing Pool Member JSON Configuration](#)

View the JSON configuration of the pool member.

[Creating TCP, PING, TLS-HELLO, and UDP Health Monitors](#)

Create a health monitor that uses low-level network probes to check whether pool members respond to connection attempts.

[Creating HTTP and HTTPS Health Monitors](#)

Create a health monitor that uses HTTP or HTTPS probes to check the availability of pool members based on response codes and path checks.

[Editing Health Monitors](#)

Update the name, max retries, or interval of an existing health monitor.

[Deleting Health Monitors](#)

Permanently delete a health monitor you no longer need.

[Viewing Health Monitor JSON Configuration](#)

View the JSON configuration of the health monitor.

Creating Pools

Create pools to define which pool members (your backend servers) the load balancer sends traffic to. You also select how the load balancer distributes traffic between these pool members.

Prerequisites

- You have a load balancer.
- **Optional:** If you want to encrypt connections to pool members with TLS, ensure the following:
    - You have created a certificate secret for TLS encryption and an authentication secret (CA) for validating connections to pool members.

      For more information, see [Key Manager](#).
    - To customize the default cipher suites, make sure you understand how cipher suites work and their impact on security.

      For more information, see [TLS Cipher Suite Catalog](#).

Procedure

1. Choose **Networking & Loadbalancing** > **Load Balancers**.
2. Choose the load balancer to which you want to add a pool.
3. Choose **New Pool**.

   A dialog box opens.
4. Enter a name.
5. **Optional:** Enter a description.

6. To define how the load balancer distributes traffic across the pool members, select a load balancing algorithm. You have the following options:

Load Balancing Algorithm	Description
LEAST_CONNECTIONS	The load balancer tracks active connections and forwards new requests to the pool member with the fewest open connections.
ROUND_ROBIN	The load balancer forwards each new request to the next pool member in line, distributing traffic evenly across all pool members. Use this method if your pool members have similar capacity and performance.

7. Select the protocol for routing traffic to pool members. You have the following options:
    - HTTP
    - PROXY
    - TCP
    - UDP
8. **Optional:** To ensure that traffic from the same client consistently reaches the same pool member, select a session persistence type.

    For each listener, you can only configure session persistence on its default pool. If you use other pools (for example, with an L7 policy), they follow the persistence settings of the default pool. You have the following options:

Session Persistence Type	Description
SOURCE_IP	The load balancer uses the client's IP address to keep all requests from the same client directed to the same pool member.
HTTP_COOKIE	The load balancer creates a cookie when it receives the first client request and uses it to route all subsequent requests from the same client to the same pool member.
APP_COOKIE	The load balancer uses a cookie created by the backend application to route all requests carrying the same cookie value to the same pool member.

9. **Optional:** To set this pool as the default for a listener, select the listener.
10. **Optional:** To encrypt connections to pool members with TLS, select the **Use TLS** checkbox.

    You can only attach a TLS-enabled pool to a TLS-terminated HTTPS listener.

    a. Select the certificate secret.

    b. Select the authentication secret (CA).

    c. Adjust the default TLS cipher suites attached to the listener.
11. **Optional:** Enter tags to organize or classify the pool.
12. Choose **Save**.

Results

- The activity is recorded as an event in the audit log.

Next Steps

- Creating TCP, PING, TLS-HELLO, and UDP Health Monitors
- Creating HTTP and HTTPS Health Monitors
- Creating Pool Members

Related Information

Audit

Editing Pools

Update the name, description, algorithm, session persistence type, or tags of an existing pool.

Prerequisites

- **Optional:** If you want to encrypt connections to pool members with TLS, ensure the following:
  - You have created a certificate secret for TLS encryption and an authentication secret (CA) for validating connections to pool members.

    For more information, see Key Manager.
  - To customize the default cipher suites, make sure you understand how cipher suites work and their impact on security.

    For more information, see TLS Cipher Suite Catalog.

Procedure

1. Choose **Networking & Loadbalancing** > **Load Balancers**.
2. Choose the load balancer that contains the pool you want to edit.
3. From the gear dropdown menu for the pool, choose **Edit**.

   A dialog box opens.
4. Update the fields you want to change.
5. Choose **Save**.

Results

- The activity is recorded as an event in the audit log.

Related Information

Audit

Deleting Pools

Permanently delete a pool you no longer need.

Procedure

1. Choose **Networking & Loadbalancing** > **Load Balancers** .
2. Choose the load balancer that contains the pool you want to delete.
3. From the gear dropdown menu for the pool, choose **Delete**.
4. Confirm the dialog box.

Results

- The pool is deleted, along with all associated pool members and health monitors, if any.
- The activity is recorded as an event in the audit log.

Related Information

Audit

Viewing Pool JSON Configuration

View the JSON configuration of the pool.

Procedure

1. Choose **Networking & Loadbalancing** > **Load Balancers** .
2. Choose the load balancer that has pool.
3. From the gear dropdown menu for the pool, choose **JSON**.

   A dialog box opens and shows the pool JSON configuration.

Creating Pool Members

Create a new pool member that serves the traffic behind a load balancer. Each pool member is indicated by the IP address and port that it uses for traffic.

Prerequisites

- You have a load balancer.

  For more information, see Creating Load Balancers.

- You have a pool.

  For more information, see Creating Pools.

- You have security groups that allow dynamic health checks.

  Load balancers undergo scaling or migration, which results in changes to health-check IPs. To avoid service interruptions, ensure that your security groups for pool members allow access from the full subnet of the load balancer.

Example

If your load balancer uses the subnet 10.180.1.0/24 and your pool members listen on port 1080, the security group has a TCP-INGRESS rule allowing 10.180.1.0/24 access to port 1080.

For more information, see Security Groups.

Procedure

1. Choose Networking & Loadbalancing > Load Balancers.
2. Choose the load balancer.
3. Choose the pool.
4. Choose New Member.

   A dialog box opens.
5. Enter a name.
6. Enter the IPs (IP address and protocol port) of the server where you want to route traffic.

   Do not add a pool member with the IP address of another load balancer in the same network.

   ### Remember

   If you add an IP from an existing server as a pool member and later change its IP address, the new IP will not be updated automatically. To update the IP, you must remove the pool member and then add it again.
7. **Optional:** Set the pool member as backup member.

   Backup members will not receive traffic unless all other pool members are down.
8. **Optional:** Adjust the weight for the new pool member.

   A weight between 0 and 100 determines how much traffic the pool member will handle. The default weight is 1.

   - To deactivate the pool member, set the weight to 0 (no traffic will be routed to it).
   - To increase the traffic for the pool member, enter a weight between 1 and 100. Higher values will result in more traffic for the member.

     ### Note

     Weights are uniformly mapped from the 0–255 range to the 0–100 range in the backend. This means that, for example, pool members with a weight of 1 and 2 will both be mapped to a weight of 1.
9. **Optional:** Enter tags to organize or classify the pool member.
10. Choose Save.

Results

- The activity is recorded as an event in the audit log.

Related Information

Audit

Editing Pool Members

Update the name, IPs, weight, or tags of an existing pool member.

Procedure

1. Choose **Networking & Loadbalancing** > **Load Balancers**.
2. Choose the load balancer.
3. Choose the pool.
4. From the gear dropdown menur for the pool member, choose **Edit**.
5. Enter a name.
6. Enter the IPs (IP address and protocol port) of the server where you want to route traffic.

   Do not add a pool member with the IP address of another load balancer in the same network.

   → Remember

   If you add an IP from an existing server as a pool member and later change its IP address, the new IP will not be updated automatically. To update the IP, you must remove the pool member and then add it again.

7. **Optional:** Set the pool member as backup member.

   Backup members will not receive traffic unless all other pool members are down.
8. **Optional:** Adjust the weight for the new pool member.

   A weight between 0 and 100 determines how much traffic the pool member will handle. The default weight is 1.

   - To deactivate the pool member, set the weight to 0 (no traffic will be routed to it).
   - To increase the traffic for the pool member, enter a weight between 1 and 100. Higher values will result in more traffic for the member.

     i Note

     Weights are uniformly mapped from the 0–255 range to the 0–100 range in the backend. This means that, for example, pool members with a weight of 1 and 2 will both be mapped to a weight of 1.

9. **Optional:** Enter tags to organize or classify the pool member.
10. **Optional:** To enable the pool member for traffic routing, select the **Admin State Up** checkbox.
11. Choose **Save**.

Results

- The activity is recorded as an event in the audit log.

Related Information

Audit

Deleting Pool Members

Delete pool members that you no longer need.

Procedure

1. Choose **Networking & Loadbalancing** > **Load Balancers**.
2. Choose the load balancer that has pools.
3. Choose the pool.
4. From the gear dropdown menu for the pool member, choose **Delete**.

Results

- The activity is recorded as an event in the audit log.

Related Information

Audit

Viewing Pool Member JSON Configuration

View the JSON configuration of the pool member.

Procedure

1. Choose **Networking & Loadbalancing** > **Load Balancers**.
2. Choose the load balancer that has pools.
3. Choose the pool.
4. From the gear dropdown menu for the pool member, choose **JSON**.

   A dialog box opens and shows the pool member JSON configuration.

Creating TCP, PING, TLS-HELLO, and UDP Health Monitors

Create a health monitor that uses low-level network probes to check whether pool members respond to connection attempts.

Prerequisites

- You have at least one running server.

  For more information, see Servers.

- You have a load balancer.

  For more information, see Creating Load Balancers.

- You have a pool.

  For more information, see Creating Pools.

Context

Health monitors regularly check whether pool members are available and responsive. Based on the selected probe type, they test either basic connectivity or application-level behavior. If a member fails to respond as expected, it is marked as offline and removed from traffic distribution.

Use TCP, PING, TLS-HELLO, or UDP-CONNECT health monitors when you only need to check that a pool member accepts and responds to connections. These types do not inspect application-level content.

Procedure

1. Choose Networking & Loadbalancing > Load Balancers.
2. Choose the load balancer.
3. Choose the pool to which you want to add a health monitor.
4. Choose New Health Monitor.

   A dialog box opens.
5. Enter a name.
6. Select the probe type:
   - PING
   - TCP
   - TLS-HELLO
   - UDP-CONNECT
7. Enter the maximum number retries before marking the pool members offline.
8. Enter the time interval in seconds between sending health check probes to pool members.
9. **Optional:** Enter tags to organize or classify the health monitor.
10. Choose Save.

Results

- The activity is recorded as an event in the audit log.

Next Steps

Make sure that your security groups or firewall rules allow traffic from the health monitors' IP addresses to the pool members.

Related Information

[Audit](#)
[Security Groups](#)

Creating HTTP and HTTPS Health Monitors

Create a health monitor that uses HTTP or HTTPS probes to check the availability of pool members based on response codes and path checks.

Prerequisites

- You have at least one running server.

  For more information, see Servers.
- You have a load balancer.

For more information, see Creating Load Balancers.

- You have a pool.

  For more information, see Creating Pools.

Context

Health monitors regularly check whether pool members are available and responsive. Based on the selected probe type, they test either basic connectivity or application-level behavior. If a member fails to respond as expected, it is marked as offline and removed from traffic distribution.

Use HTTP or HTTPS health monitors to check application-level health. You can define expected status codes, request paths, and methods to verify that pool members serve valid application responses.

Procedure

1. Choose **Networking & Loadbalancing** > **Load Balancers**.
2. Choose the load balancer.
3. Choose the pool.
4. Choose **New Health Monitor**.

   A dialog box opens.
5. Enter a name.
6. Select the probe type:
   - **HTTPS**
   - **HTTP**
7. Enter the maximum number of retries before marking the pool members offline.
8. Enter the time interval in seconds between sending health check probes to pool members.
9. Select the HTTP method:

Method	Description
CONNECT	Initiates a two-way communication tunnel (typically for TLS)
DELETE	Requests deletion of the specified resource
GET	Retrieves data without modifying the server
HEAD	Retrieves headers only, no body
OPTIONS	Returns the communication options for the target
PATCH	Applies partial modifications to a resource
POST	Sends data, typically to create resources
PUT	Replaces the resource at the target URL

10. Enter the expected HTTP status codes.

    You can use the following formats:
    - A single value, such as 200

- A comma-separated list, such as 200, 202
- A range, such as 200–204

11. Enter the URL path the health monitor requests.

    It must begin with a forward slash (/).
12. **Optional:** Enter tags to organize or classify the health monitor.
13. Choose **Save**.

Results

- The activity is recorded as an event in the audit log.

Next Steps

Make sure that your security groups or firewall rules allow traffic from the health monitors' IP addresses to the pool members.

Related Information

[Audit](#)
[Security Groups](#)

Editing Health Monitors

Update the name, max retries, or interval of an existing health monitor.

Procedure

1. Choose **Networking & Loadbalancing** > **Load Balancers**.
2. Choose the load balancer.
3. Choose the pool.
4. From the gear dropdown menu for the health monitor, choose **Edit**.

   A dialog box opens.
5. Enter a name.
6. Enter the maximum number of failed probes sent to the pool members before marking such pool members offline.
7. Enter the time interval in seconds between sending health check probes to pool members.
8. Choose **Save**.

Results

- The activity is recorded as an event in the audit log.

Related Information

[Audit](#)

Deleting Health Monitors

Permanently delete a health monitor you no longer need.

Procedure

1. Choose Networking & Loadbalancing > Load Balancers.
2. Choose the load balancer.
3. Choose the pool.
4. From the gear dropdown menu for the health monitor, choose **Delete**.

Results

- The activity is recorded as an event in the audit log.

Related Information

[Audit]

Viewing Health Monitor JSON Configuration

View the JSON configuration of the health monitor.

Procedure

1. Choose Networking & Loadbalancing > Load Balancers.
2. Choose the load balancer that has pools.
3. Choose the pool.
4. From the gear dropdown menu for the health monitor, choose **JSON**.

   A dialog box opens and shows the health monitor JSON configuration.

Domain Name System

Manage domain name system (DNS) zones and records.

[Domain Name System Record Types]

Get an overview of the domain name system (DNS) record types.

[Sharing Domain Name System Zones with other Projects]

Share domain name system (DNS) zones with other projects, but keep the records of the current project visible only to that project.

[Deleting Domain Name System Zones]

Delete domain name system (DNS) zones that you no longer need.

[Creating Domain Name System Records]

Create domain name system (DNS) records to define domain mappings, mail server settings, and other network configurations.

[Editing Domain Name System Records]

Update the content, description, and time-to-live of an existing domain name system (DNS) record.

[Deleting Domain Name System Records]

Delete domain name system (DNS) records that you no longer need.

Domain Name System Record Types

Get an overview of the domain name system (DNS) record types.

Record Type	**Description**
A	Internet Protocol version 4 (IPv4) address record Maps a domain to an IPv4 address.
AAAA	Internet Protocol version 6 (IPv6) address record Maps a domain to an IPv6 address.
CAA	Certification authority authorization record Specifies which certificate authorities can issue secure sockets layer (SSL) or transport layer security (TLS) certificates for the domain.
CERT	Certificate record Stores cryptographic certificates, including those used for SSL, TLS, and secure/multipurpose internet mail extensions (S/MIME).
CNAME	Canonical name record Creates an alias from one domain name to another.
HTTPS	HTTPS service binding record A specialized SVCB record for HTTP services. Allows clients to discover supported protocols, ports, and security-related parameters before connecting.
PTR	Pointer record Maps an internet protocol address to a domain name for reverse DNS lookups.
SPF	Sender policy framework record Specifies which mail servers are authorized to send e-mails on behalf of a domain, helping to prevent e-mail spoofing and phishing.
SRV	Service locator record Defines the hostname and port for specific services.
SVCB	Service binding record Provides connection details for a service endpoint, such as supported protocols, ports, alternative endpoints, and IP hints.
TXT	Text record Stores human-readable text data, commonly used for domain verification and security policies.

Sharing Domain Name System Zones with other Projects

Share domain name system (DNS) zones with other projects, but keep the records of the current project visible only to that project.

Prerequisites

- You have the `dns_webmaster` role.

  For more information, see Authorizations.

- The DNS zone you want to share with another project must not already be shared with your project. After sharing, the target projects cannot share the zone with additional projects.

Procedure

1. Choose **Networking & Loadbalancing** > **DNS**.
2. From the gear dropdown menu for the zone, choose **Share this Zone**.

   A dialog box opens.
3. Enter a target project ID.
4. Chose **Share**.

Deleting Domain Name System Zones

Delete domain name system (DNS) zones that you no longer need.

Prerequisites

- You have the `dns_webmaster` role.

  For more information, see Authorizations.

Procedure

1. Choose **Networking & Loadbalancing** > **DNS**.
2. From the gear dropdown menu for the zone, choose **Delete**.

Creating Domain Name System Records

Create domain name system (DNS) records to define domain mappings, mail server settings, and other network configurations.

Prerequisites

- You have the `dns_webmaster` role.

  For more information, see Authorizations.

- You know the supported record types.

  For more information, see Domain Name System Record Types.

Procedure

1. Choose **Networking & Loadbalancing** > **DNS**.
2. Choose the zone.
3. Choose **Create New**.

   A dialog box opens.
4. Select the record type.
5. **Optional:** Enter a name other than the zone name itself.
6. Based on the selected record type, enter the corresponding content.

   Example

   For A records, enter the IPv4 address.

   For CNAME records, enter the canonical name.
7. **Optional:** Enter a description.
8. **Optional:** Adjust the time-to-live (TTL) for the record, in seconds.
9. Choose **Create**.

Related Information

[Floating IPs]

Editing Domain Name System Records

Update the content, description, and time-to-live of an existing domain name system (DNS) record.

Prerequisites

- You have the `dns_webmaster` role.

  For more information, see [Authorizations].

Procedure

1. Choose **Networking & Loadbalancing** > **DNS**.
2. Choose the zone.
3. From the gear dropdown menu for the record, choose **Edit**.

   A dialog box opens.
4. Based on the record type, enter the corresponding content.

   Example

   For A records, enter the IPv4 address.

   For CNAME records, enter the canonical name.
5. **Optional:** Enter a description.
6. **Optional:** Adjust the time-to-live (TTL) for the record, in seconds.

7. Choose **Update**.

Deleting Domain Name System Records

Delete domain name system (DNS) records that you no longer need.

Prerequisites

- You have the `dns_webmaster` role.

  For more information, see Authorizations.

Procedure

1. Choose **Networking & Loadbalancing** > **DNS**.
2. Choose the zone.
3. From the gear dropdown menu for the record, choose **Delete**.

API Access

Manage API authentication and configuration using endpoint parameters.

Viewing API Endpoint Parameters
View the API endpoint parameters required for authentication and configuration. These parameters help you securely access and interact with the API.

Downloading the OpenStack RC File
Download the OpenStack RC file that provides API endpoint parameters for authentication and configuration.

Viewing API Endpoint Parameters

View the API endpoint parameters required for authentication and configuration. These parameters help you securely access and interact with the API.

Procedure

1. Choose **API Access** > **API Endpoints for Clients**.
2. View the following parameters along with their current values, excluding sensitive information such as user name and password:

Parameter	Description
`OS_AUTH_URL`	The authentication URL for your API endpoint.
`OS_IDENTITY_API_VERSION	OS_AUTH_VERSION`	The version of the identity API.
`OS_REGION_NAME`	The name of the region.
`OS_USER_DOMAIN_ID	OS_USER_DOMAIN_NAME`	The ID and name of the user domain.

Parameter	Description
OS_PROJECT_DOMAIN_ID | OS_PROJECT_DOMAIN_NAME	The ID and name of the project domain.
OS_PROJECT_ID | OS_PROJECT_NAME	The ID and name of the project.
OS_USERNAME	The user name for authentication. **Note** The UI does not display the user name for authentication.
OS_PASSWORD	The password for authentication. **Note** The UI does not display the password for authentication.

Downloading the OpenStack RC File

Download the OpenStack RC file that provides API endpoint parameters for authentication and configuration.

Procedure

1. Choose API Access > API Endpoints for Clients.
2. Download the file. You have the following options:

Option	Description

Download Openstack RC File	openrc-*<domain>*-*<project>* file for Linux and MacOS:
	Sample Code
	`export OS_AUTH_URL="https://identity-3.<region>.cloud.sap/v3"`
	`export OS_IDENTITY_API_VERSION="3"`
	`export OS_PROJECT_NAME="<your project>"`
	`export OS_PROJECT_DOMAIN_NAME="<your project domain>"`
	`export OS_USERNAME="<your user name>"`
	`export OS_USER_DOMAIN_NAME="<your user domain>"`
	`echo "Please enter your OpenStack Password: "`
	`read -sr OS_PASSWORD_INPUT`
	`export OS_PASSWORD=$OS_PASSWORD_INPUT`
	`export OS_REGION_NAME="<region>"`

Download Openstack RC File for Windows PowerShell	openrc-*<domain>*-*<project>*.ps1 file for Windows:
	Sample Code
	`$env:OS_AUTH_URL="https://identity-3.<region>.cloud.sap/v3"`
	`$env:OS_IDENTITY_API_VERSION="3"`
	`$env:OS_PROJECT_NAME="<your project>"`
	`$env:OS_PROJECT_DOMAIN_NAME="<your project domain>"`
	`$env:OS_USERNAME="<your user name>"`
	`$env:OS_USER_DOMAIN_NAME="<your user domain>"`

Option	Description

	`$Password = Read-Host -Prompt "OpenStack User Password?" -AsSecureString`
	`$env:OS_PASSWORD = [Runtime.InteropServices.Marshal]::PtrToStringAuto([Runtime.InteropS`
	`$env:OS_REGION_NAME="<region>"`

3. Save the file to a location of your choice.

Related Information

[Sourcing the OpenStack RC File](#)
[Creating the OpenStack RC File for Password-Based Authentication](#)
[Creating the OpenStack RC File for Token-Based Authentication](#)

Capacity and Metrics

Track resource usage and quotas. Explore metrics and audit logs to monitor and analyze your cloud environment.

[Resource Management](#)
Manage and monitor your usage of resources such as compute, network, and storage.

[Metrics](#)
Explore the available metrics and understand how they're categorized with labels. Use the metrics dashboard to analyze the data.

[Audit](#)
Get an overview of audit logs, including how to view and analyze events, and understand the attributes recorded for each event.

Resource Management

Manage and monitor your usage of resources such as compute, network, and storage.

[Checking the Resource Quota](#)
Monitor your resource usage for compute, DNS, network, and storage resources.

Checking the Resource Quota

Monitor your resource usage for compute, DNS, network, and storage resources.

Prerequisites

- You have the `resource_viewer` role.

  For more information, see [Authorizations](#).

Procedure

1. Choose Capacity and Metrics > Resource Management.
2. Choose one of the following tab pages for the resource type you want to check:

Tab Page	Resource Type
Compute	Cores Server Instances Members per Server Group RAM Server Groups Flavors
DNS	Recordsets per Zone Zones
Network	BGP VPNs Floating IPs Networks Ports RBAC Policies Routers Security Group Rules Security Groups Subnet Pools Subnets Trunks Health Monitors L7 Policies Listeners Load Balancers Pool Members Pools
Storage	Capacity Snapshots Volumes Images

Tab Page	Resource Type
	◦ Share Capacity ◦ Share Networks ◦ Share Snapshot Capacity ◦ Share Snapshots ◦ Shares ◦ Snapmirror Capacity
PAYG Availability	◦ Compute ◦ Block Storage ◦ Object Storage ◦ Shared Filesystem Storage

Metrics

Explore the available metrics and understand how they're categorized with labels. Use the metrics dashboard to analyze the data.

[Metrics Catalog](#)
Get an overview of available metrics.
[Labels for Metrics](#)
Get an overview of the labels that categorize and organize metrics.
[Logging On to the Metrics Dashboard](#)
The metrics dashboard is an adapted Prometheus expression browser. Log on to the metrics dashboard to explore metrics, view time series, and perform ad-hoc queries by using the full capabilities of PromQL's query syntax.

Metrics Catalog

Get an overview of available metrics.

Metric Name	Description	Metric Source	Resource
`snmp_f5_ltmVirtualServStatClientBytesIn`	Total number of bytes received by the specified virtual server from the client side.	OpenStack	Load Balancers
`snmp_f5_ltmVirtualServStatClientBytesOut`	Total number of bytes sent by the specified virtual server	OpenStack	Load Balancers

Metric Name	Description	Metric Source	Resource
	to the client side.		
`snmp_f5_ltmVirtualServStatClientCurConns`	Current number of connections from the client side to the specified virtual server.	OpenStack	Load Balancers
`snmp_f5_ltmVirtualServStatClientMaxConns`	Maximum number of connections from the client side to the specified virtual server.	OpenStack	Load Balancers
`snmp_f5_ltmVirtualServStatClientTotConns`	Total number of connections from the client side to the specified virtual server.	OpenStack	Load Balancers
`snmp_f5_ltmVirtualServStatTotRequests`	Total number of requests processed by the specified virtual server.	OpenStack	Load Balancers
`snmp_f5_ltmVirtualServStatName`	Name of the virtual server.	OpenStack	Load Balancers
`ssh_nat_limits_use`	Number of dynamic NAT translations for a virtual router.	OpenStack	Routers
`ssh_nat_limits_miss`	Number of dynamic NAT translations that could not be created due to reaching the limit.	OpenStack	Routers
`snmp_asr_ifHCInOctets`	Total number of octets received on the interface, including	OpenStack	Routers

Metric Name	Description	Metric Source	Resource
	framing characters.		
snmp_asr_ifHCOutOctets	Total number of octets transmitted from the interface, including framing characters.	OpenStack	Routers
vrops_virtualmachine_config_hardware_memory_kilobytes	Configured memory for the virtual machine in kilobytes.	VMware vRealize Operations	Server Instances
vrops_virtualmachine_cpu_contention_ratio	CPU contention as a percentage over a 20-second collection interval.	VMware vRealize Operations	Server Instances
vrops_virtualmachine_cpu_demand_ratio	CPU demand as a percentage of the provisioned capacity.	VMware vRealize Operations	Server Instances
vrops_virtualmachine_cpu_usage_ratio	Percentage of allocated CPU that was used.	VMware vRealize Operations	Server Instances
vrops_virtualmachine_cpu_usage_average_mhz	CPU usage in megahertz.	VMware vRealize Operations	Server Instances
vrops_virtualmachine_cpu_wait_summation_miliseconds	CPU time spent waiting for I/O operations.	VMware vRealize Operations	Server Instances
vrops_virtualmachine_cpu_workload_percentage	CPU workload percentage. The maximum threshold is 80%, and the minimum threshold is 20%.	VMware vRealize Operations	Server Instances

Metric Name	Description	Metric Source	Resource
vrops_virtualmachine_datastore_outstanding_io_requests	Number of outstanding I/O operations.	VMware vRealize Operations	Server Instances
vrops_virtualmachine_disk_usage_average_kilobytes_per_second	Disk usage rate in kilobytes per second.	VMware vRealize Operations	Server Instances
vrops_virtualmachine_diskspace_gigabytes	Provisioned disk space in gigabytes.	VMware vRealize Operations	Server Instances
vrops_virtualmachine_diskspace_virtual_machine_used_gigabytes	Disk space used by virtual machine files in gigabytes.	VMware vRealize Operations	Server Instances
vrops_virtualmachine_guest_os_full_name_info	Name of the guest operating system.	VMware vRealize Operations	Server Instances
vrops_virtualmachine_guest_tools_version_info	Version of the guest operating system tools.	VMware vRealize Operations	Server Instances
vrops_virtualmachine_memory_active_ratio	Percentage of total memory that is actively used.	VMware vRealize Operations	Server Instances
vrops_virtualmachine_memory_activewrite_kilobytes	Active memory writes in kilobytes.	VMware vRealize Operations	Server Instances
vrops_virtualmachine_memory_balloning_ratio	Percentage of total memory reclaimed through ballooning.	VMware vRealize Operations	Server Instances
vrops_virtualmachine_memory_consumed_kilobytes	Total consumed memory in kilobytes.	VMware vRealize Operations	Server Instances
vrops_virtualmachine_memory_contention_ratio	Percentage of memory contention.	VMware vRealize Operations	Server Instances
vrops_virtualmachine_memory_usage_average	Average memory	VMware vRealize	Server Instances

Metric Name	Description	Metric Source	Resource
	usage of the virtual machine.	Operations	
vrops_virtualmachine_network_data_received_kilobytes_per_second	Rate of data received by the virtual machine in kilobytes per second.	VMware vRealize Operations	Server Instances
vrops_virtualmachine_network_data_transmitted_kilobytes_per_second	Rate of data transmitted by the virtual machine in kilobytes per second.	VMware vRealize Operations	Server Instances
vrops_virtualmachine_network_packets_dropped_rx_number	Number of received packets dropped during the collection interval.	VMware vRealize Operations	Server Instances
vrops_virtualmachine_network_packets_dropped_tx_number	Number of transmitted packets dropped during the collection interval.	VMware vRealize Operations	Server Instances
vrops_virtualmachine_network_packets_rx_number	Number of packets received during the performance interval.	VMware vRealize Operations	Server Instances
vrops_virtualmachine_network_packets_tx_number	Number of packets transmitted during the performance interval.	VMware vRealize Operations	Server Instances

Metric Name	Description	Metric Source	Resource
vrops_virtualmachine_network_usage_average_kilobytes_per_second	Total data transmitted and received for all NIC instances between the virtual machine and host.	VMware vRealize Operations	Server Instances
vrops_virtualmachine_number_vcpus_total	Total number of virtual CPUs assigned to the virtual machine.	VMware vRealize Operations	Server Instances
vrops_virtualmachine_oversized	Indicates whether the virtual machine is oversized.	VMware vRealize Operations	Server Instances
vrops_virtualmachine_oversized_vcpus	Number of virtual CPUs that should be removed from an oversized virtual machine.	VMware vRealize Operations	Server Instances
vrops_virtualmachine_oversized_memory	Amount of memory (in kilobytes) that should be removed from an oversized virtual machine.	VMware vRealize Operations	Server Instances
vrops_virtualmachine_runtime_connectionstate		VMware vRealize Operations	Server Instances
vrops_virtualmachine_runtime_powerstate	Power state of the virtual machine: 1 if powered on, 0 if powered off, -1 if unknown.	VMware vRealize Operations	Server Instances
vrops_virtualmachine_summary_ethernetcards		VMware vRealize	Server Instances

Metric Name	Description	Metric Source	Resource
		Operations	
vrops_virtualmachine_swapin_memory_kilobytes	Amount of swap-in memory in kilobytes.	VMware vRealize Operations	Server Instances
vrops_virtualmachine_swapped_memory_kilobytes	Amount of memory swapped out in kilobytes.	VMware vRealize Operations	Server Instances
vrops_virtualmachine_undersized	Indicates whether the virtual machine is undersized.	VMware vRealize Operations	Server Instances
vrops_virtualmachine_undersized_vcpus	Number of virtual CPUs to remove from an undersized virtual machine.	VMware vRealize Operations	Server Instances
vrops_virtualmachine_undersized_memory	Amount of memory (in kilobytes) to remove from an undersized virtual machine.	VMware vRealize Operations	Server Instances
vrops_virtualmachine_virtual_disk_average_read_miliseconds	Average read latency of the virtual disk in milliseconds (total latency = kernel latency + device latency).	VMware vRealize Operations	Server Instances
vrops_virtualmachine_virtual_disk_average_write_miliseconds	Average write latency of the virtual disk in milliseconds (total latency = kernel latency + device latency).	VMware vRealize Operations	Server Instances
vrops_virtualmachine_virtual_disk_outstanding_io	Number of outstanding I/O requests.	VMware vRealize Operations	Server Instances

Metric Name	Description	Metric Source	Resource
vrops_virtualmachine_virtual_disk_read_kilobytes_per_second	Read rate from the virtual disk in kilobytes per second.	VMware vRealize Operations	Server Instances
vrops_virtualmachine_virtual_disk_write_kilobytes_per_second	Write rate to the virtual disk in kilobytes per second.	VMware vRealize Operations	Server Instances
openstack_compute_instances_gauge	Number of instances per project based on virtual machine state.	OpenStack	Server Instances
openstack_compute_stuck_instances_count_gauge	Number of instances stuck for more than 15 minutes.	OpenStack	Server Instances
openstack_compute_stuck_instances_max_duration_gauge	Maximum duration an instance has been stuck (more than 15 minutes).	OpenStack	Server Instances
openstack_stuck_volumes_count_gauge	Total number of volumes that are stuck.	OpenStack	Block Storage
openstack_stuck_volumes_max_duration_gauge	Maximum duration a volume has been stuck (more than 15 seconds).	OpenStack	Block Storage
openstack_volumes_count_gauge	Number of volumes per project.	OpenStack	Block Storage
openstack_volumes_size_gauge	Total size of volumes per project.	OpenStack	Block Storage
openstack_snapshots_count_gauge	Number of snapshots per project.	OpenStack	Block Storage

Metric Name	Description	Metric Source	Resource
openstack_snapshots_size_gauge	Total size of snapshots per project.	OpenStack	Block Storage
openstack_manila_shares_count_gauge	Number of file shares per project.	OpenStack	File System Storage
openstack_manila_shares_size_gauge	Total size of file shares per project.	OpenStack	File System Storage
openstack_manila_snapshot_count_gauge	Number of file share snapshots per project.	OpenStack	File System Storage
openstack_manila_snapshot_size_gauge	Total size of file share snapshot per project.	OpenStack	File System Storage
openstack_manila_replicas_count_gauge	Number of file share replicas per project.	OpenStack	File System Storage
openstack_manila_share_connection_count	Number of active connections to a given file share.	OpenStack	File System Storage
openstack_manila_share_total_bytes	Total available space in bytes for a file share.	OpenStack	File System Storage
openstack_manila_share_used_bytes	Space used in bytes by a file share.	OpenStack	File System Storage
openstack_manila_share_available_bytes	Available space in bytes for a file share.	OpenStack	File System Storage
openstack_manila_share_used_percentage	Percentage of used space in a file share.	OpenStack	File System Storage
openstack_manila_share_snapshot_reserved_bytes	Space reserved in bytes for creating	OpenStack	File System Storage

Metric Name	Description	Metric Source	Resource
	snapshots of a file share.		
openstack_manila_share_snapshot_used_bytes	Space used in bytes for snapshots of a file share.	OpenStack	File System Storage
openstack_manila_share_snapshot_available_bytes	Available space in bytes for snapshots of a file share.	OpenStack	File System Storage
openstack_manila_share_inode_files_total	Total number of inode files in a file share.	OpenStack	File System Storage
openstack_manila_share_inode_files_used	Number of used inode files in a file share.	OpenStack	File System Storage
openstack_manila_share_inode_files_used_percentage	Percentage of used inode files in a file share.	OpenStack	File System Storage
openstack_manila_share_total_ops	Total number of operations performed on a file share.	OpenStack	File System Storage
openstack_manila_share_read_ops	Total number of read operations on a file share.	OpenStack	File System Storage
openstack_manila_share_write_ops	Total number of write operations on a file share.	OpenStack	File System Storage
openstack_manila_share_read_latency_microseconds	Read latency in microseconds for a file share.	OpenStack	File System Storage
openstack_manila_share_write_latency_microseconds	Write latency in microseconds for a file share.	OpenStack	File System Storage
openstack_manila_share_read_throughput	Read throughput in bytes per	OpenStack	File System Storage

Metric Name	Description	Metric Source	Resource
	second for a file share.		
openstack_manila_share_write_throughput	Write throughput in bytes per second for a file share.	OpenStack	File System Storage
openstack_manila_snapmirror_labels	SnapMirror replication status information.	OpenStack	Data Replication
openstack_manila_snapmirror_lag_time	Time lag in seconds for SnapMirror data transfer.	OpenStack	Data Replication
openstack_manila_snapmirror_last_transfer_duration	Duration in seconds of the last SnapMirror data transfer.	OpenStack	Data Replication
openstack_audit_events_counter	Total number of audit log events emitted for a project.	Hermez	Audit
limes_project_quota	Assigned quota for a resource in a project.	Limes	Resource Management
limes_project_usage	Current usage of a resource in a project.	Limes	Resource Management
limes_project_commitment_min_expires_at	Expiration date of a committed resource.	Limes	Resource Management
openstack_glance_image_gauge	Details of images, including status, visibility, and size.	OpenStack	Images
openstack_glance_image_created_gauge	Number of images created in the last 24 hours.	OpenStack	Images

Metric Name	Description	Metric Source	Resource
openstack_glance_image_deleted_gauge	Number of images deleted in the last 24 hours.	OpenStack	Images
openstack_barbican_certificate_expiration_date	Expiration date of certificates per project.	OpenStack	Key Manager
openstack_barbican_containers_count_gauge	Total number of secrets containers per project, categorized by type and status.	OpenStack	Key Manager
openstack_barbican_secrets_count_gauge	Total number of secrets per project, categorized by type and status.	OpenStack	Key Manager
keppel_manifests_with_unclean_vuln_status_count	Number of container images per repository with a vulnerability status other than "Clean".	Keppel	Container Image Registry
objectstore_bytes_received_total	Total number of bytes received per bucket.	CEPH	Object Storage
objectstore_bytes_sent_total	Total number of bytes sent per bucket.	CEPH	Object Storage
objectstore_read_ops_total	Total number of read operations per bucket.	CEPH	Object Storage
objectstore_write_ops_total	Total number of write operations per bucket.	CEPH	Object Storage
objectstore_capacity_usage_bytes	Capacity usage per bucket.	CEPH	Object Storage

Metric Name	Description	Metric Source	Resource
`objectstore_usage_objects`	Object usage per bucket.	CEPH	Object Storage
`objectstore_client_io_latency_seconds_avg`	Client I/O latency (GET+PUT) – average over 5 minutes, in seconds.	CEPH	Object Storage
`objectstore_upload_latency_milliseconds_avg`	Upload object latency (PUT+POST) – average over 5 minutes, in milliseconds.	CEPH	Object Storage
`objectstore_download_latency_milliseconds_avg`	Download object latency (GET) – average over 5 minutes, in milliseconds.	CEPH	Object Storage
`objectstore_delete_latency_milliseconds_avg`	Delete object latency – average over 5 minutes, in milliseconds.	CEPH	Object Storage
`objectstore_client_error_rate`	Client error rate (4xx) – per second, measured per second, per bucket, and per error type.	CEPH	Object Storage
`objectstore_server_error_rate`	Server / service error rate (5xx) – measured per second, per bucket, and per error type.	CEPH	Object Storage

Labels for Metrics

Get an overview of the labels that categorize and organize metrics.

Label Name	Description	Resource
`project_id`	The unique identifier (UUID) of the project associated with the resource.	Load Balancers Server Instances Block Storage File System Storage Audit Key Manager Auto Scaling Container Images Object Storage
`lb_id`	The unique identifier (UUID) of the load balancer.	Load Balancers
`listener_id`	The unique identifier (UUID) of the listener.	Load Balancers
`network_id`	The unique identifier (UUID) of the network.	Load Balancers
`virtualmachine`	The name and unique identifier (UUID) of the virtual machine.	Server Instances
`vcenter`	The name of the vCenter where the virtual machine is running.	Server Instances
`hostsystem`	The name of the bare metal hypervisor node where the virtual machine is running.	Server Instances
`flavor_name`	The name of the flavor used by the virtual machine.	Server Instances
`host`	The name of the host associated with the resource, such as a virtual machine, volume, or snapshot.	Server Instances Block Storage
`vm_state`	The current state of the virtual machine.	Server Instances
`status`	The current state of the resource, such as a volume, snapshot, key, or file system share.	Block Storage File System Storage Key Manager
`attach_status`	The current attachment status of the volume.	Block Storage
`availability_zone`	The name of the availability zone in which the volume is located.	Block Storage
`volume_type_id`	The unique identifier (UUID) of the volume type associated with the volume.	Block Storage
`id`	The unique identifier (UUID) of the file system storage.	File System Storage

Label Name	Description	Resource
`share_type_id`	The unique identifier (UUID) of the file system share type.	File System Storage
`share_id`	The unique identifier (UUID) of the file system share.	File System Storage
`volume`	The unique identifier (UUID) of the volume on the backend filer.	File System Storage
`vserver`	The unique identifier (UUID) of the vServer to which the file system share belongs on the backend filer.	File System Storage
`relationship_id`	The unique identifier of the SnapMirror relationship.	Data Replication
`relationship_type`	The type of SnapMirror relationship, such as mirror or vault.	Data Replication
`relationship_status`	The current operational status of the SnapMirror relationship.	Data Replication
`healthy`	Indicates whether the SnapMirror relationship is considered healthy (`true` or `false`).	Data Replication
`schedule`	The name of the transfer schedule associated with the SnapMirror relationship.	Data Replication
`source_cluster`	The name of the source cluster in the SnapMirror relationship.	Data Replication
`source_vserver`	The name of the source vServer.	Data Replication
`source_volume`	The name of the source volume.	Data Replication
`destination_cluster`	The name of the destination cluster in the SnapMirror relationship.	Data Replication
`destination_node`	The node in the destination cluster that hosts the SnapMirror relationship.	Data Replication
`destination_vserver`	The name of the destination vServer.	Data Replication
`destination_volume`	The name of the destination volume.	Data Replication
`destination_project_id`	The unique identifier (UUID) of the project that owns the destination share.	Data Replication
`destination_share_id`	The unique identifier (UUID) of the destination file system share.	Data Replication
`destination_share_name`	The name of the destination file system share.	Data Replication
`last_transfer_type`	The type of the last data transfer performed in the SnapMirror relationship.	Data Replication

Label Name	Description	Resource
`last_transfer_error`	The error message (if any) from the last SnapMirror data transfer.	Data Replication
`action`	The activity performed on the target resource, such as an update.	Audit
`outcome`	The result of the action performed on the resource, such as success, failure, or unknown.	Audit
`target_type_uri`	The type of the target resource, such as compute/server for a virtual machine.	Audit
`service`	The service performing the change, such as compute.	Audit Resource Management
`resource`	The specific resource within a service, such as cores.	Resource Management
`certificate_name`	The current name of the certificate.	Key Manager
`secret_name`	The current name of the secret type.	Key Manager
`container_name`	The current name of the container.	Key Manager
`asset`	The type of asset associated with the resource, such as nfs-shares.	Auto Scaling
`from_state`	The state of the operation before the transition.	Auto Scaling
`to_state`	The state of the operation after the transition.	Auto Scaling
`account`	The name of container image registry account.	Container Image Registry
`repo`	The name of repository within the container image registry account.	Container Image Registry
`vuln_status`	The vulnerability status of the container image, such as `Pending` for newly pushed images or `Unsupported` if the image can't be scanned for vulnerabilities.	Container Image Registry
`bucket`	The container name.	Object Storage

Logging On to the Metrics Dashboard

The metrics dashboard is an adapted Prometheus expression browser. Log on to the metrics dashboard to explore metrics, view time series, and perform ad-hoc queries by using the full capabilities of PromQL's query syntax.

Prerequisites

- You have the `monitoring_viewer` role.

For more information, see Authorizations.

Procedure

1. Choose **Capacity and Metrics** > **Metrics**.
2. Choose **Open Maia Dashboard**.

   The dashboard opens.

   ### Example

   ```
   https://maia.<region>.cloud.sap/<domain>
   ```

Audit

Get an overview of audit logs, including how to view and analyze events, and understand the attributes recorded for each event.

→ Tip

Regularly backup your audit logs to comply with your organization's data retention policies.

Audit Log Event Catalog
Get an overview of audit log events.

Audit Log Event Attributes
Get an overview of the attributes recorded in audit logs for each event.

Viewing Audit Logs
View and analyze audit logs to monitor events. An event represents a user or system action.

Audit Log Event Catalog

Get an overview of audit log events.

This catalog provides a representative selection of audit log events but is not comprehensive:

Service	Observer Type	Target Type	Related Links
Key Management	service/data/security/keymanager	data/security/keymanager/container data/security/keymanager/order data/security/keymanager/secret data/security/keymanager/secret/acl data/security/keymanager/secret/metadata data/security/keymanager/secret/payload	Key Manager
Load Balancing	service/loadbalancer	network/loadbalancer network/loadbalancer/l7policy/rule network/loadbalancer/l7policies	Load Balancers

Service	Observer Type	Target Type	Related Links
		network/loadbalancer/pool network/loadbalancer/healthmonitor network/loadbalancer/listener network/loadbalancer/pools/member	
Network	service/network	network/bgpvpn network/bgvpn/router-association network/firewalls network/floatingip network/loadbalancers network/network network/port network/quotas network/router network/security-group network/security-group-rule network/subnet	Networking and Loadbalancing
Compute	service/compute	compute/keypair compute/server compute/server/interface compute/server/metadata compute/server/tags compute/server/volume-attachment compute/quotas	Compute
	service/resources	service/compute/ram/quota	
Identity Management	service/security	service/security/account/user data/security/account/user data/security/account/group data/security/credential data/security/domain data/security/project	Authorizations
Authoritative Domain Name System	service/service/dns	service/dns/zone service/dns/zone/recordset	Domain Name System
Server Images	service/storage/image	storage/image/image storage/image/image/file	Server Images and Server Image Snapshots

Service	Observer Type	Target Type	Related Links
		storage/image/image/members	
File Storage	service/storage/share	storage/share/security-service storage/share/share storage/share/share-network storage/share/snapshot storage/share/share-replica	[File System Storage]
Block Storage	service/storage/volume	storage/volume/attachment storage/volume/snapshot storage/volume/types storage/volume/volume	[Volumes and Volume Snapshots - Block Storage]
Container Images	service/docker-registry	docker-registry/account docker-registry/account/repository/manifest docker-registry/account/repository/tag docker-registry/project-quota	[Container Image Registry]
Auto Scaling	service/autoscaling	data/security/project	[Auto Scaling]

Audit Log Event Attributes

Get an overview of the attributes recorded in audit logs for each event.

Attribute	Description
Observer Type	Specifies the type of resource observing the event. **Example** `service/security`
Action	Describes the type of activity performed. Action identifiers follow a hierarchical structure, starting with a top-level classification (such as create, read, or update) and optionally followed by a more specific sub-classification. **Example** `update/add/floatingip`
Target Type	Specifies the type of resource targeted by the event. **Example** `compute/server`
Target ID	Represents the ID of the targeted resource.

Attribute	Description
	Example server ID
Initiator ID	Represents the ID of the initiating resource. **Example** user ID
Initiator Name	**Optional:** Provides a descriptive name of the initiating resource.
Initiator Type	Specifies the type of the initiating resource. **Example** `/data/security/account/user`
Outcome	Indicates the result of the activity. Optionally, a response code can be included. **Example** • `failure` • `success` • `unknown`

Viewing Audit Logs

View and analyze audit logs to monitor events. An event represents a user or system action.

Prerequisites

- You have the `audit_viewer` role.

  For more information, see Authorizations.

- You're familiar with the event attributes in audit logs and understand their meanings.

  For more information, see Audit Log Event Attributes.

Procedure

1. Choose Capacity and Metrics > Audit.

   The audit log shows the events.
2. **Optional:** Filter events based on their time of occurrence or specific event attributes.

Troubleshooting

Having issues? We want to help.

If you can't find a solution by searching the SAP Help Portal, report an issue for application component `GCID-SCI-001` in SAP for Me. Our SAP Cloud Infrastructure support specialists regularly monitor issues assigned to this application component and process them promptly.

[Error Message Reference for File System Storage Operations]

Get an overview of predefined error codes and their corresponding messages used in file system storage operations to support your troubleshooting.

Related Information

SAP for Me

SAP for Me Online Help

Error Message Reference for File System Storage Operations

Get an overview of predefined error codes and their corresponding messages used in file system storage operations to support your troubleshooting.

Code	Error Message
001	`An unknown error occurred.`
002	`No storage could be allocated for this share request. Trying again with a different size or share type may succeed.`
003	`Driver does not expect share-network to be provided with current configuration.`
004	`Could not find an existing share server or allocate one on the share network provided. You may use a different share network, or verify the network details in the share network and retry your request. If this doesn't work, contact your administrator to troubleshoot issues with your network.`
005	`An 'active' replica must exist in 'available' state to create a new replica for share.`
006	`Share has no replica with 'replica_state' set to 'active'.`
007	`No storage could be allocated for this share request, AvailabilityZone filter didn't succeed.`
008	`No storage could be allocated for this share request, Capabilities filter didn't succeed.`
009	`No storage could be allocated for this share request, Capacity filter didn't succeed.`
010	`No storage could be allocated for this share request, Driver filter didn't succeed.`
011	`No storage could be allocated for this share request, IgnoreAttemptedHosts filter didn't succeed.`
012	`No storage could be allocated for this share request, Json filter didn't succeed.`

Code	Error Message
013	No storage could be allocated for this share request, Retry filter didn't succeed.
014	No storage could be allocated for this share request, ShareReplication filter didn't succeed.
015	Share Driver failed to extend share, The share status has been set to extending_error. This action cannot be re-attempted until the status has been rectified. Contact your administrator to determine the cause of this failure.
016	No storage could be allocated for this share request, CreateFromSnapshot filter didn't succeed.
017	Share Driver has failed to create the share from snapshot. This operation can be re-attempted by creating a new share. Contact your administrator to determine the cause of this failure.
018	Share Driver refused to shrink the share. The size to be shrunk is smaller than the current used space. The share status has been set to available. Please select a size greater than the current used space.
019	Share Driver does not support shrinking shares. Shrinking share operation failed.
020	Failed to grant access to client. The client ID used may be forbidden. You may try again with a different client identifier.
021	Failed to grant access to client. The access level or type may be unsupported. You may try again with a different access level or access type.
022	Share driver has failed to setup one or more security services that are associated with the used share network. The security service may be unsupported or the provided parameters are invalid. You may try again with a different set of configurations.
023	Share Driver failed to create share due to a security service authentication issue. The security service user has either insufficient privileges or wrong credentials. Please check your user, password, ou and domain.
024	No default share type has been made available. You must specify a share type for creating shares.
025	Share Driver failed to create share because a security service has not been added to the share network used. Please add a security service to the share network.
026	Share Driver failed to delete the share. Create a support case.
027	No storage could be allocated for this share request, AffinityFilter filter didn't succeed.
028	No storage could be allocated for this share request, AntiAffinityFilter filter didn't succeed.

Code	Error Message
029	Share Driver failed to create share server on share network due to port limit exceeded. You may increase quota of network ports and retry your request. If this doesn't work, contact your administrator to troubleshoot issues with your network.
030	Share Driver failed to create share server on share network due no more free IP addresses in the neutron subnet. You may free some IP addresses in the subnet or create a new subnet/share network. If this doesn't work, contact your administrator to troubleshoot issues with your network.

Related Information

[File System Storage]

[SAP for Me]

[SAP for Me Online Help]